

Learning to Allocate: Capacity Constraints and Kidney Discard in Deceased Donor Allocation

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Each year thousands of recovered kidneys are discarded because repeated offer rejections accumulate cold ischemia time until the organ is no longer viable. This report investigates transplant center capacity constraints, particularly surgeon availability, as a driver of offer rejection. Using offer data from SRTR, a Bayesian mixed-effects acceptance model, a donor-level discard model, and a center capacity model based on monthly active surgeon counts are estimated. Several capacity-aware acceptance models are compared; a two-tower neural network performs best, suggesting that match quality and capacity behave as complementary inputs rather than substitutes. The two-tower model is embedded in an allocation simulation to quantify how surgeon capacity limits discard reduction, with further analysis of urban-rural differences in capacity pressure. For organs at high risk of discard, an RL allocation policy is explored, optimizing for acceptance while preserving ex ante waitlist fairness. Capacity-aware acceptance modeling reduces simulated discard by roughly 29 percentage points relative to a non-capacity-aware acceptance model.

Keywords: kidney allocation, organ discard, mechanism design, cold ischemia time, reinforcement learning

JEL Codes: D47, I11, I18, C61

1 Introduction

Kidney transplantation is the best treatment for end-stage kidney disease, offering longer survival and higher quality of life than dialysis. Yet the gap between supply and demand remains severe. In 2023, a record 28,142 kidney transplants were performed in the United States, while over 140,000 candidates were listed on the waiting list at some point during the year, and more than 4,000 candidates died while waiting (Organ Procurement and Transplantation Network (OPTN) and Scientific Registry of Transplant Recipients (SRTR) 2024). Recognizing the urgency of the problem, the OPTN Expeditious Task Force recently set a goal of increasing total U.S. transplants to 60,000 by 2026 (Miller et al. 2024).

A central obstacle to reaching this target is not organ supply alone but organ utilization. Approximately 22–25% of recovered deceased donor kidneys are discarded each year, a rate that has risen since 2021 (Mohan et al. 2023). Published estimates suggest that roughly half of all discarded kidneys are lost not because of poor organ quality but because of repeated offer rejections that accumulate cold ischemia time until the organ is no longer viable (Mohan et al. 2023). In effect, thousands of transplantable kidneys are wasted each year by the process through which they are offered.

Figure 1 illustrates the causal structure of this process. When a deceased donor kidney becomes available, the UNOS match-run algorithm generates a ranked list of candidates based on priority rules incorporating factors such as KDPI, EPTS, blood type compatibility, and geographic proximity. The organ is then offered sequentially down this list. At each step, the decision to accept or reject involves a triad of agents: the patient, the transplant surgeon, and the transplant center. Each agent evaluates the offer through an optimal stopping lens, weighing the current organ against the expected value of waiting for a potentially better future offer. If the organ is rejected, it moves to the next candidate on the list. With each rejection, cold ischemia time grows. If the list is exhausted or the organ has been out of storage for too long, it is discarded.

Deceased Donor Kidney Allocation — Causal Diagram

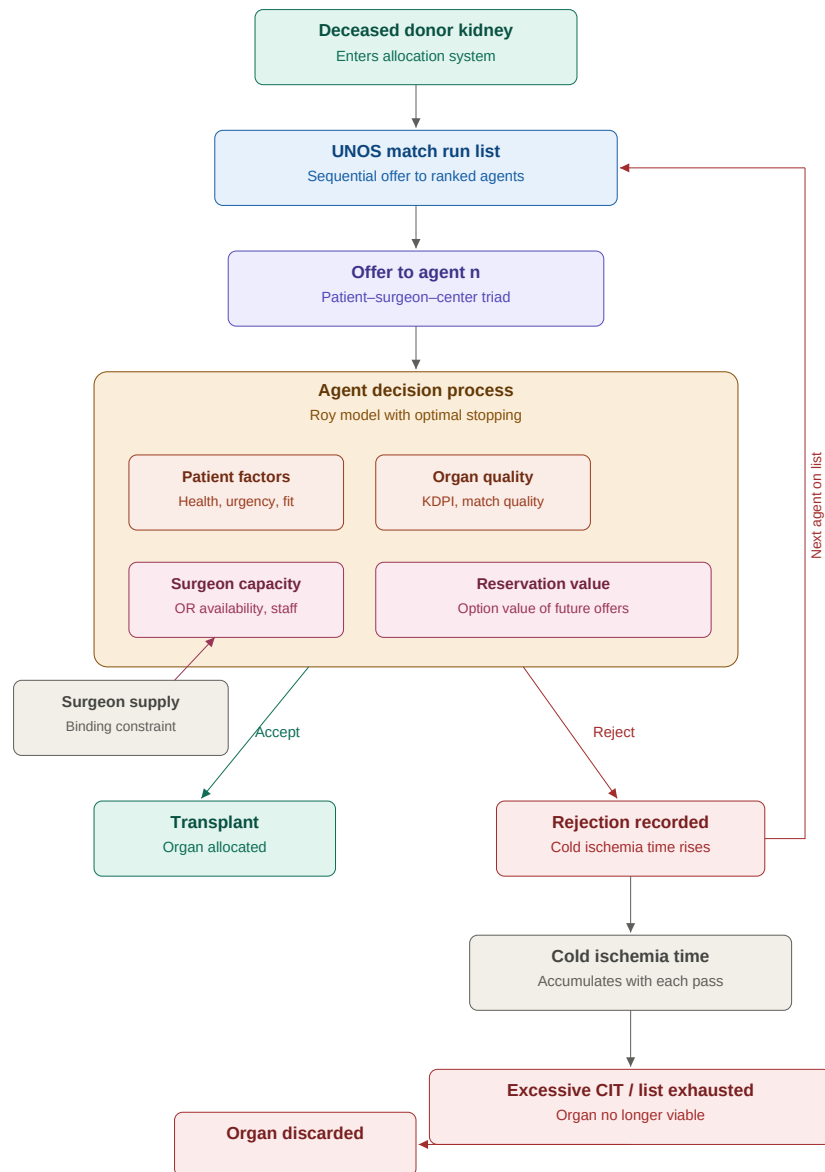


Figure 1: Causal structure of the deceased donor kidney allocation process. An organ enters the match-run list and is offered sequentially to ranked agents (patient–surgeon–center triads). Each agent’s decision incorporates organ quality, patient factors, surgeon capacity, and the reservation value of future offers. Repeated rejections accumulate cold ischemia time (CIT), ultimately leading to discard if the organ is not accepted.

The existing literature on organ allocation has made substantial progress in modeling

the match quality channel of this decision: how organ and patient characteristics interact to determine acceptance and post-transplant outcomes (Agarwal et al. 2025, 2021). Far less attention has been paid to the operational channel. Transplant surgery requires not just a willing patient and a suitable organ but an available surgeon and an open operating room. If a center’s surgical team is fully committed when an offer arrives, the offer will be declined regardless of how good the match is. Preliminary evidence from the Scientific Registry of Transplant Recipients suggests that surgeon availability is positively associated with transplant volume and negatively associated with organ non-use (Miller et al. 2024), but this relationship has not been modeled at the individual offer level.

This report investigates transplant center capacity constraints, particularly surgeon availability, as a driver of offer rejection and kidney discard. The analysis proceeds in several stages. First, a Bayesian mixed-effects acceptance model is estimated on individual-level SRTR offer data, with random intercepts for patients, donors, and centers, to establish a baseline prediction of offer-level acceptance. Second, a donor-side model is fitted predicting whether a given donor’s kidney is likely to be continually rejected until discard. Third, a separate center capacity model and a capacity to transplant volume regression are estimated using novel SRTR data on the number of unique surgeons active at each transplant center per month. Fourth, the logic of the two models is combined in several capacity-aware acceptance models that separates rejections attributable to match quality from those attributable to operational bottlenecks. Fifth, the capacity-aware model is embedded in an allocation simulation to quantify how much surgeon capacity constraints limit the system’s ability to reduce discard. The simulation is run with and without the capacity-aware predictions, isolating the contribution of surgical bottlenecks to the discard rate. The simulation also logs capacity events by center type, enabling an analysis of urban-rural differences in capacity pressure and workload intensity. Finally, for organs predicted to be at high risk of discard, a reinforcement learning policy is explored that reranks candidates using a learned distance metric over donor-candidate features, optimizing for acceptance while preserving ex ante

waitlist fairness by selectively activating.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on allocation mechanism design, fairness, staffing constraints, and the discard rate problem. Section 3 describes all datasets used in the analysis. Section 4 presents the empirical models and the simulation framework. Section 5 reports results. Section 6 concludes.

2 Literature Review

2.1 Mechanism Design and Allocation Analysis

A central insight of recent research on organ allocation is that outcomes cannot be understood by examining medical compatibility alone—patients and clinicians respond strategically to the incentives embedded in allocation rules. Agarwal et al. (2025) make this point most clearly. In their joint model of acceptance decisions and post-transplant survival, the authors show that rational rejection of viable organs is a major driver of discards, and that the current U.S. mechanism produces far fewer life-years from transplant than would a planner-optimized allocation. By simulating counterfactual policies in equilibrium, they demonstrate that substantial welfare gains are attainable without increasing donor supply. Their analysis reveals a fundamental design gap: discards are not simply logistical failures but equilibrium consequences of the existing policy environment. Among the current literature, this paper most directly informs the framework of the present study—linking behavioral acceptance and clinical outcomes, and using policy simulation to evaluate alternative mechanisms.

Agarwal et al. (2021) provide complementary evidence that allocation rules shape patient decisions in dynamic and forward-looking ways. Modeling each offer as an optimal-stopping choice under uncertainty, they show that small changes in prioritization can meaningfully alter discard rates, wait times, and welfare in steady-state equilibrium. Their results underscore that reforms cannot be evaluated outside of agent behavior: even if supply, KDPI,

and match quality remain fixed, waitlist incentives change how transplants are accepted or deferred. Together, the 2025 and 2021 papers establish that any credible assessment of allocation policy must incorporate behavioral responses, not just clinical feasibility.

This behavioral perspective builds on a broader econometric tradition. Hotz and Miller (1993) introduce the conditional choice probability method that allows dynamic decision problems with large state spaces to be estimated without repeatedly solving the full dynamic program. Arcidiacono and Ellickson (2011) formalize practical approaches to dynamic discrete choice estimation and counterfactual simulation across a range of applied settings. These methods have become standard in the transplant literature because they render equilibrium policy evaluation computationally feasible.

2.2 Allocation Fairness Frameworks

Where much of this work emphasizes efficiency and utilization, a separate literature highlights the central role of fairness. Dale et al. (2024) survey current U.S. allocation scoring systems—such as MELD, LAS, KDPI, and EPTS—and document inconsistencies, subgroup disparities, and a persistent emphasis on short-term mortality over equitable access. Their review argues that existing scoring frameworks often embed implicit value judgments without transparency, and that predictive modeling and allocation should be disentangled. This critique underscores the need for designs that improve utilization without reinforcing inequity or myopic prioritization.

Parallel work in public health and medical ethics emphasizes that allocation legitimacy depends on recognizable fairness principles. Emanuel et al. (2020) describe how allocation systems must balance multiple ethical goals simultaneously—maximizing total benefit, treating individuals equally, giving priority to the worst-off, and preserving instrumental value. Persad et al. (2009) similarly argue for “multiprinciple” allocation frameworks rather than single-metric rules, noting that attempts to optimize only one dimension—such as survival gains—often generate ethically unacceptable disparities. These principles directly motivate

the approach of this paper, which evaluates mechanisms using a loss function that prioritizes efficiency while selectively activating to preserve the expected chances of receiving an organ in aggregate.

Survey evidence suggests that these fairness concerns are not merely theoretical. Lee et al. (2021) find that the general public systematically prefers allocation rules that balance maximizing lives saved with treating individuals fairly, often rejecting first-come-first-served or purely utilitarian policies. Huseynov (2020) similarly shows that most respondents support prioritizing both the sickest and those most likely to benefit—again reflecting a multiprinciple perspective aligned with ethical theory.

Behavioral fairness studies within economics reinforce this insight. Andreoni et al. (2020) show that individuals exhibit context-dependent fairness preferences: before outcomes are known, most people prefer *ex ante* fairness—equal chances of receiving an organ—but once uncertainty is resolved, many shift to *ex post* fairness—favoring the sicker or worse-off. These preference reversals highlight how legitimacy depends on both procedural and distributive fairness, and that mechanisms must be evaluated on more than aggregate survival statistics.

Taken together, this literature provides the foundation for evaluating alternative kidney allocation mechanisms. Structural models show that discards and welfare losses are endogenous consequences of current rules, not inevitable features of medical scarcity. Ethical and survey research demonstrates that public support depends on fairness principles recognizable to stakeholders, not just efficiency. The present study builds on Agarwal et al. (2025) by adopting their equilibrium evaluation perspective, but extends the policy space to consider mechanisms that reduce discard while maintaining equitable access. In doing so, it bridges the gap between structural efficiency analysis and normative concerns about fairness and legitimacy in organ allocation.

2.3 System Staffing Constraints

A smaller but meaningful literature examines how staffing limitations constrain transplant throughput, independent of organ availability. One dimension involves the clinical workforce needed to evaluate donors and recipients, perform surgery, and manage post-operative care. For example, analysis of the transplant nephrology workforce in the United States documents persistent shortages in trained nephrology professionals and highlights recruitment and pipeline challenges that may restrict capacity for expanding kidney transplantation services (Crowley et al. 2020).

Surgical staffing presents a related bottleneck. The transplantation surgery pipeline has been identified as constrained, with concerns about insufficient numbers of junior surgeons entering transplant fellowships relative to projected clinical demand (Quillin and Cortez 2022). Because most abdominal transplant surgeons are trained first in general surgery, concerns about long-run general surgeon shortfalls imply downstream constraints on transplant workforce capacity.

Preliminary empirical evidence supports the idea that surgeon availability is positively associated with transplant volume. A poster presented by the Scientific Registry of Transplant Recipients (SRTR), titled *The Surgical Workforce and Its Relationship to Kidney Transplant Numbers and Nonuse*, models the relationship between monthly kidney transplants and the number of unique surgeons performing transplant procedures in the same period (Miller et al. 2024). The poster suggests that greater surgeon availability predicts higher transplant throughput and lower organ non-use, implying that staffing shortages may convert otherwise viable organs into discards. While this evidence is not yet peer-reviewed, it underscores the importance of considering system-level constraints—such as surgeon capacity—when evaluating mechanisms intended to reduce discard.

These staffing constraints intersect with a well-documented urban-rural divide in physician supply. Although 20% of the U.S. population lives in rural areas, only about 10% of physicians practice there, and rural counties had 1.6 times fewer primary care physicians

per capita than urban counties as of 2019 (Arredondo et al. 2023). Recent data indicate that this gap is widening: between 2017 and 2023, rural areas experienced a net loss of 11% of family physicians nationwide, even as rural populations grew. The shortage extends beyond primary care to specialty and surgical fields, where the concentration of training programs and practice opportunities in urban academic centers leaves rural regions structurally underserved.

In transplantation specifically, Axelrod et al. (2008) document that rural residents are 8–15% less likely to be waitlisted and 10–20% less likely to receive a kidney transplant than urban residents, with the gap consistent across all OPTN regions. A scoping review of barriers to kidney transplantation in rural and remote areas identifies workforce shortages, distance to transplant centers, and limited access to specialty evaluation as recurring themes from both provider and patient perspectives (Watters et al. 2023). While these studies focus on patient access rather than center-level surgical capacity, they suggest that rural transplant centers may face compounding disadvantages: fewer surgeons, lower baseline volume, and greater difficulty absorbing surges in organ offers. To our knowledge, no prior study has directly characterized the urban-rural distribution of transplant surgeon supply at the center level. This is a gap that the SRTR surgeon data can begin to address.

2.4 The Organ Discard Rate Problem

Kidney transplantation remains critically constrained not only by organ supply, but by high rates of organ discard. Recent data indicate that approximately 104,000 individuals are currently on the U.S. transplant waitlist, including roughly 90,000 waiting for kidneys, yet only 27,279 kidney transplants were performed in 2024 according to OPTN. This is a system that directly affects the lives of tens of thousands of patients, but still suffers from substantial inefficiency. One of the most visible symptoms of this inefficiency is the persistently high organ discard rate. Estimates from the National Institutes of Health suggest an overall discard rate near 22% across organs (Mohan et al. 2023).

A principal driver of discard is repeated rejection of viable kidneys until they are no longer clinically usable. Published estimates attribute between 47.5% and 58.5% of discarded kidneys to continued refusal of offers until organ quality deteriorates below a usable threshold (Mohan et al. 2023). In response, organ procurement organizations have increasingly employed “skipping” policies—preemptively bypassing high-priority candidates in order to quickly offer organs to recipients deemed more likely to accept.

However, this intervention has been controversial. Investigative reporting has alleged that skipping creates opportunities for special interest influence in allocation, and statistical analyses indicate that skipped organs disproportionately go to white and Asian male recipients (Rosenthal et al. 2025). Moreover, recent evidence shows that a large proportion of skips occur long before an organ approaches its expiration window, calling into question whether skipping is used solely as an emergency measure. Despite the rapid growth of skipping—now accounting for an estimated 20% of kidney transplants—the national discard rate has continued to rise (Mohan et al. 2023). These patterns suggest that skipping has not achieved its intended purpose and may exacerbate inequity without meaningfully improving utilization.

Taken together, the organ discard problem illustrates structural inefficiencies and highlights the need for allocation mechanisms that reduce waste while preserving fairness. A key question for policy design is whether alternative incentive-compatible mechanisms can reduce discard without requiring discretionary overrides of formal priority rules.

2.5 Reinforcement Learning for Allocation Problems

Reinforcement learning provides a natural framework for sequential decision problems under uncertainty, and has been applied to a growing range of healthcare and resource allocation contexts (Yu et al. 2023). Li et al. (2024) apply deep Q-learning to ventilator triage during the COVID-19 pandemic, demonstrating that learned policies can outperform heuristic-based protocols on both efficiency and equity dimensions. Within the organ transplantation literature, several papers have explored RL-based allocation strategies. Bertsimas et al. (2013)

formulate kidney allocation as an optimization problem with explicit fairness constraints, providing the operations research foundation for subsequent learning-based approaches. Xu et al. (2021) (OrganITE) propose a representation-learning framework for individualized donor-recipient matching that estimates counterfactual outcomes for unobserved matches. More recently, Naqvi et al. (2025) introduce an RL framework that learns daily allocation decisions to balance net survival benefit against waiting penalties.

This study departs from prior work in two ways. First, the policy is parameterized as a learnable transformation matrix on candidate-organ feature differences, preserving the structural interpretability of a Mahalanobis-style distance metric while allowing the policy to learn which feature dimensions matter most for acceptance. Second, the simulation environment incorporates surgeon capacity constraints through the two-tower acceptance model that produces capacity-aware acceptance probabilities at the candidate level, a channel that prior RL allocation work has not addressed. The result is an allocation policy that responds not only to organ-patient match quality but also to the operational state of the receiving center.

3 Data

This study draws on five datasets obtained from the Scientific Registry of Transplant Recipients (SRTR) through a standard data and an enhanced request, as well as one dataset from the census bureau. The first three—offer-level records, patient registrations, and donor records—are standard SRTR research files widely used in the transplant economics literature. The latter two—center-level surgeon supply counts and metro center designations—provide operational and geographic detail not available in the core files. The census source is 2020 data on state-level urban and rural populations. Together, these datasets support both the offer-level acceptance model and the center capacity constraint model described in Section 4.

3.1 Offer Data (PTR)

The Potential Transplant Recipient (PTR) file is the primary modeling dataset. Each row represents a single offer of a deceased-donor kidney to a specific waitlisted candidate, recorded in the order generated by the UNOS match-run algorithm. The file covers all kidney offers made in the United States from January 2015 through December 2022, totaling approximately 164 million offer records.

Key variables include `DONOR_ID` (linking to the donor file), `WL_ID_CODE` (linking to the patient file), `LISTING_CTR` (the transplant center receiving the offer), `PTR_ROW_ORDER` (the candidate’s sequential position in the match run), `MATCH_DATE` (the date the match run was generated), and `OFFER_ACCEPT` (the binary accept/reject outcome). The file also contains geographic coordinates of both the donor hospital and the transplant center, offer notification and response timestamps, and coded refusal reasons.

The acceptance rate in the PTR data is low, typically between 2% and 5% depending on the time window, reflecting the sequential nature of the allocation process: most offers are declined before a kidney finds an accepting recipient. For estimation, we restrict the sample to offers made between September and December 2022, with an August 2022 warmup window used to construct lagged features.

3.2 Patient Data (KIDPAN)

The KIDPAN file contains one row per waitlist registration for kidney and/or pancreas transplant candidates. It includes approximately 1.12 million registrations spanning the full history of the UNOS waitlist through December 2022.

For each registration, the file records patient demographics (age at listing, sex, blood type, race/ethnicity, BMI, state of residence), immunological profile (HLA typing, calculated panel reactive antibody or cPRA), medical status (primary diagnosis, diabetes status, dialysis status and date, functional status), and waitlist history (listing date, days waiting, activation date, removal code). For patients who received a transplant, the file additionally contains

transplant-date variables, donor characteristics as recorded at the time of transplant, HLA mismatch counts, cold ischemia time, and post-transplant outcomes including graft survival, serum creatinine at follow-up intervals, and patient survival.

The patient file is merged onto the offer data via `WL_ID_CODE` in a many-to-one join: each candidate appears in many offers but has a single patient record. Variables used in the acceptance model include `INIT_AGE`, `ABO`, `GENDER`, `DIAB`, `ON_DIALYSIS`, `INIT_CPRA`, `END_CPRA`, `BMI_TCR`, `INIT_DATE` (used to compute wait time), and `REGION`.

3.3 Donor Data (STAR)

The STAR donor file contains one row per deceased donor, with approximately 88,500 donors over the study period. We restrict to donors with at least one kidney recovered (`NUM_KI_RECOV` > 0).

Each record includes donor demographics (age, sex, blood type, BMI, race/ethnicity), cause and circumstances of death (including DCD status), clinical labs (terminal creatinine, BUN, liver enzymes), organ quality indices (KDPI, KDRI), serological testing results, donor medical history (diabetes, hypertension, substance use), kidney-specific biopsy and perfusion data, and organ disposition codes indicating whether each kidney was transplanted or discarded.

The donor file is merged onto the offer data via `DONOR_ID` in a many-to-one join: each donor generates many offers but has a single donor record. Key variables for the acceptance model include `AGE_DON`, `KDPI`, `CREAT_DON`, `ABO_DON`, `GENDER_DON`, `HIST_DIABETES_DON`, `HIST_HYPERTENS_DON`, and `NON_HRT_DON`. The `discarded` indicator is used as the training label for the organ-level discard prediction model described in Section 4.3.

3.4 Center-Level Surgeon Supply

The center-level surgeon supply data were received from SRTR via a standard data request. These data consist of three panel datasets at the monthly, quarterly, and yearly levels. Each records, for a given transplant center and time period, the number of unique transplant surgeons who performed at least one kidney transplant procedure at that center during the period.

For example, the monthly file contains one row per center-month pair with a count of distinct surgeons active in that month. The quarterly and yearly files aggregate analogously. These counts serve as the primary input to the center capacity constraint model: a center with fewer active surgeons in a given month faces a tighter operational constraint and may be more likely to reject offers that it would otherwise accept. The monthly panel is used for model estimation; the quarterly and yearly panels provide robustness checks and longer-horizon capacity measures.

3.5 Center-Level Urban-Rural Designation

The center-level urban-rural designation data were also received from SRTR via a standard data request. This dataset provides a mapping from each transplant center to an urban or rural classification based on the geographic location of the center.

This variable is used as a covariate in the capacity constraint model. Urban centers typically have larger surgical teams and higher baseline throughput but may also face greater demand, while rural centers may have fewer surgeons but more stable capacity utilization. The urban-rural distinction allows the capacity model to account for structural differences in how surgeon supply translates into effective transplant capacity across center types.

3.6 Census Urban-Rural State Data

State-level urban and rural population data were obtained from the U.S. Census Bureau’s 2020 Decennial Census urban-rural classification (U.S. Census Bureau 2022). The dataset reports, for each of the 50 states and the District of Columbia, total within-state urban and rural populations along with their corresponding percentages, where rural is defined as the population residing outside of areas classified as urban under the post-2020 Census criteria. The within-state urban populations sum to 265,149,027 across the 50 states and DC, exactly matching the Census Bureau’s published national total.

This variable is used in the state-level analysis of the simulation outputs to examine whether the percentage of a state’s population residing in rural areas correlates with simulated capacity pressure or simulated discard rate. State of each offer is identified from the first two letters of the four-letter center code, which encode the standard two-letter U.S. postal abbreviation. The state rurality measure provides an externally-defined population-based characterization of each state that is independent of the simulation pipeline, allowing the rural-state analysis to triangulate with the center-level urban-rural analysis described in Section 3.5.

4 Methodology

4.1 Allocation Environment

We model deceased-donor kidney allocation as a sequential decision problem in which organs arrive over time and are offered to candidates on the waitlist until accepted or discarded. For each arriving organ o , the planner must determine an ordered sequence of candidate offers. Acceptance decisions and post-transplant outcomes are stochastic and depend on both donor and candidate characteristics.

To evaluate counterfactual allocation policies, we embed the learned allocation rule within

a simulation environment that incorporates pre-estimated acceptance (choice) and survival (outcome) models, as estimated in Agarwal et al. (2025). These models map donor–candidate pairs to acceptance probabilities and post-transplant life-years from transplant (LYFT), respectively.

4.2 Markov Decision Process Formulation

We formulate allocation as a Markov Decision Process (MDP) (Sutton and Barto 2018). At each decision step t , the system is in state s_t , which summarizes: (i) the characteristics of the currently available organ o_t ; (ii) the current waitlist snapshot, represented by candidate feature vectors $x_{ii \in \mathcal{W}_t}$; and (iii) contextual information such as the offer position and calendar time.

An action is implicitly defined by the policy-induced ranking over candidates. The agent’s decision variable is the transformation matrix $\mathbf{A}$, which parameterizes a stochastic policy over candidate offers. Given $\mathbf{A}$ and state s_t , the environment selects the next candidate according to

$$\min_{x_i} d_A(x_i, o)^2 = \|A(x_i - o)\|_2^2 \quad (1)$$

and executes the offer process until acceptance or discard.

4.3 Empirical Models

This section describes the empirical models used to predict offer-level acceptance decisions in the counterfactual allocation simulations. The modeling strategy proceeds in stages: first, a baseline acceptance model estimated on SRTR offer-level data; second, a center capacity constraint model capturing operational bottlenecks; third, an ensemble combining both into a unified acceptance predictor; and fourth, an organ-level discard prediction model.

4.3.1 Notation and Data Objects

Let i index candidates (waitlist registrations), j index deceased-donor kidneys, and c index transplant centers. The estimation sample is constructed at the offer level: each observation corresponds to a single offer of kidney j to candidate i listed at center c , observed in the SRTR Potential Transplant Recipient (PTR) file.

Let:

- $x_i \in \mathbb{R}^{d_x}$ denote candidate covariates at the time of offer (age, BMI, cPRA, blood type, diabetes status, time on waitlist, etc.), drawn from the KIDPAN patient file.
- $q_j \in \mathbb{R}^{d_q}$ denote kidney/donor covariates (donor age, KDPI, terminal creatinine, donor blood type, donor sex, DCD status, etc.), drawn from the STAR donor file.
- m_{ij} denote offer-level match characteristics constructed from the pair (x_i, q_j) , including ABO identical match indicator and $\log(1 + \text{PTR_ROW_ORDER})$, which captures the candidate's position in the match-run sequence.
- $z_{ic} \in \mathbb{R}^{d_z}$ denote local market and center-level features that proxy for the option value of waiting and recent center workload. These include lagged offer volume and unique donor counts in the candidate's OPO–blood-type–wait-time cell over the prior four quarters, as well as center-specific rolling 28-day offer volume and mean KDPI for the candidate's blood type.

4.3.2 Acceptance Model: Bayesian Mixed-Effects Logistic Regression

For each offer (i, j) at center c , define the binary acceptance outcome

$$D_{ij} \in \{0, 1\},$$

where $D_{ij} = 1$ if the offer is accepted and $D_{ij} = 0$ if it is rejected. The acceptance model is a Bayesian mixed-effects logistic regression (GLMM) of the form:

$$D_{ij} \sim \text{Bernoulli}\left(\text{logit}^{-1}\left(\mathbf{w}'_{ij}\boldsymbol{\beta} + b_i^{\text{patient}} + b_j^{\text{donor}} + b_c^{\text{center}}\right)\right), \quad (2)$$

where $\mathbf{w}_{ij}$ is the concatenated fixed-effects design vector comprising:

- **Patient covariates:** age at listing (`INIT_AGE`), BMI (`BMI_TCR`), initial and end cPRA (`INIT_CPRA`, `END_CPRA`), years on waitlist (`waited_years`), patient sex and blood type indicators.
- **Donor/organ covariates:** donor age (`AGE_DON`), KDPI, terminal creatinine (`CREAT_DON`), donor sex and blood type indicators.
- **Match covariates:** ABO identical match indicator (`abo_identical`), log offer position (`log1p_ptr_row_order`).
- **Scarcity/continuation-value proxies:** log lagged 4-quarter offer volume (`log1p_lag_offers_4q`), log lagged 4-quarter unique donor count (`log1p_lag_donors_4q`), log center–blood-type 28-day offer volume (`log1p_center_abo_past_offers_28d`), center–blood-type 28-day mean KDPI (`center_abo_past_mean_kdpi_28d`). These endogenous variables work to emulate optimal stopping behavior of patients in the agentic modeling.

All continuous covariates are standardized to zero mean and unit variance prior to estimation. The random intercepts are:

$$b_i^{\text{patient}} \sim \mathcal{N}(0, \sigma_{\text{patient}}^2), \quad (3)$$

$$b_j^{\text{donor}} \sim \mathcal{N}(0, \sigma_{\text{donor}}^2), \quad (4)$$

$$b_c^{\text{center}} \sim \mathcal{N}(0, \sigma_{\text{center}}^2), \quad (5)$$

which capture unobserved heterogeneity in patient acceptance propensity, donor desirability, and center-level aggressiveness, respectively. The model is estimated via variational

Bayes using `BinomialBayesMixedGLM` in `statsmodels`, with weakly informative priors on the fixed effects (`fe_p = 5.0`) and variance components (`vcp_p = 0.5`).

The model is trained on offers from September–November 2022 and evaluated on a held-out December 2022 test set, with a one-month warmup window (August 2022) used to construct the lagged scarcity and center-level features. Offers are capped at the first 100 per donor to manage computational cost and reduce noise from late-sequence offers that are almost never accepted.

The output of this model for any offer (i, j) is a predicted probability:

$$\hat{p}_{\text{acc}}(i, j) = \text{logit}^{-1}\left(\mathbf{w}'_{ij}\hat{\boldsymbol{\beta}}\right) \in [0, 1]. \quad (6)$$

Note that at inference time on new donor–candidate pairs, predictions use only the fixed effects; the random intercepts contribute to in-sample fit but are not available for out-of-sample offers involving previously unseen patients, donors, or centers.

4.3.3 Center-Month Volume Regression

Before specifying a center-level capacity model, this study first establishes whether a meaningful capacity constraint can be detected at all in the relationship between organs offered to a center and organs that center ultimately transplants. The motivation is structural: if surgeon supply is genuinely constraining, the throughput of organs offered into transplants should depend on staffing levels, with diminishing returns to additional offers when surgeons are scarce. Miller et al. (2024) provide the methodological template for this test. Their poster regresses national-monthly transplants on the natural log of kidneys recovered, the number of unique active surgeons, the mean kidney donor risk index (KDRI), the mean cold ischemic time (CIT), and two-way interactions between log recovery and surgeon count, and between log recovery and KDRI.

This study replicates that specification at the center-month level rather than the national-

monthly level, providing finer granularity at the cost of additional idiosyncratic variation, and extends the specification with a metropolitan indicator. Let T_{ct} denote the count of kidneys transplanted at center c in month t . The volume regression takes the form:

$$\begin{aligned} T_{ct} = & \alpha_0 + \alpha_1 \log(O_{ct}) + \alpha_2 S_{ct} + \alpha_3 \overline{\text{KDRI}}_{ct} + \alpha_4 \overline{\text{CIT}}_{ct} + \alpha_5 M_c \\ & + \alpha_6 \log(O_{ct}) \cdot S_{ct} + \alpha_7 \log(O_{ct}) \cdot \overline{\text{KDRI}}_{ct} + \varepsilon_{ct}, \end{aligned} \quad (7)$$

where O_{ct} is the number of unique kidneys offered to center c in month t , S_{ct} is the number of unique transplant surgeons active at the center that month, $\overline{\text{KDRI}}_{ct}$ is the mean KDRI of offered kidneys, $\overline{\text{CIT}}_{ct}$ is the mean cold ischemic time of transplanted kidneys, and M_c is an indicator equal to 1 if the center is in a metropolitan area. The model is estimated by ordinary least squares on the full 2015–2022 panel.

The interaction $\log(O_{ct}) \cdot S_{ct}$ is the central object of interest. A positive and significant coefficient on this interaction indicates that the marginal benefit of additional offers is increasing in surgeon supply—equivalently, that surgeon supply imposes a binding constraint on the conversion of offers into transplants. The metropolitan indicator M_c captures any residual urban-rural difference in throughput that is not mediated by the included structural features. Reduced-form evidence from this regression motivates the more granular center capacity model that follows.

4.3.4 Center Capacity Constraint Model

The volume regression establishes that surgeon supply binds throughput at the aggregate center-month level. To use this insight in the offer-level acceptance prediction problem, this study estimates a separate center capacity model that classifies each center-month as either operating at or near capacity, or not, and produces a continuous probability $\hat{p}_{\text{cap}}(c, t) \in [0, 1]$ for use as a feature in the downstream ensemble.

Label construction. Because no direct ground-truth label for capacity strain exists in the data, the label is constructed behaviorally from offer-level rejection patterns. For each (center, month) cell, the rejection rate r_{ct} is computed as the share of offers received that were declined. Because center rejection rates are mechanically correlated with both offer volume and offer quality (centers that receive many low-quality offers reject most of them regardless of capacity), the rejection rate is residualized against the log of offers received and the mean KDPI of offers via OLS:

$$r_{ct} = \gamma_0 + \gamma_1 \log(O_{ct}) + \gamma_2 \overline{\text{KDPI}}_{ct} + u_{ct}. \quad (8)$$

The residual $\hat{u}_{ct}$ captures the component of rejection rate that is not explained by offer volume or offer quality, and is interpreted as excess rejection plausibly driven by operational constraints. A center-month is then flagged as “at capacity” if $\hat{u}_{ct}$ is in the top quartile within its calendar month:

$$Y_{ct}^{\text{cap}} = \mathbb{1}\{\hat{u}_{ct} \geq Q_{75}(\hat{u}_{c't} : c' \in \mathcal{C}_t)\}, \quad (9)$$

where $\mathcal{C}_t$ is the set of centers active in month t . The within-month quantile cutoff controls for any month-level shifts in baseline rejection patterns.

Specification. The capacity model is an L2-regularized logistic regression of Y_{ct}^{cap} on three features: the number of unique active surgeons S_{ct} , the center’s historical mean monthly transplant volume $\overline{T}_c^{\text{hist}}$ computed from PTR data prior to the model window, and the mean offered KDPI in the current month $\overline{\text{KDPI}}_{ct}$:

$$\text{logit}(\hat{p}_{\text{cap}}(c, t)) = \delta_0 + \delta_1 S_{ct} + \delta_2 \overline{T}_c^{\text{hist}} + \delta_3 \overline{\text{KDPI}}_{ct}. \quad (10)$$

Continuous features are standardized to zero mean and unit variance prior to estimation. The metropolitan indicator M_c is intentionally excluded from the capacity model so that urban-rural differences in capacity strain can emerge as an outcome of the simulation analysis rather than as a baked-in feature. The offers-received variable is also excluded, as it would re-introduce the mechanical correlation that the residualized label was constructed to remove.

The model is trained on center-months from August through November 2022, with December 2022 held out for evaluation, matching the window of the offer-level acceptance model. A continuous probability output is preferred over a binary classification because capacity pressure exists on a spectrum—a center running three of four surgeons faces a qualitatively different constraint than one running four of four—and a continuous signal preserves gradient information for the downstream ensemble.

4.3.5 Capacity-Aware Acceptance Architectures

Combining the offer-level acceptance signal with the center-level capacity signal into a unified acceptance predictor is itself a modeling choice. This study estimates and compares three architectures that span the spectrum from interpretable structural decomposition to flexible end-to-end learning. All three are trained against the offer-level acceptance label D_{ij} and evaluated on the same held-out test offers, allowing predictive performance and interpretability to be traded off transparently.

Sequential ensemble. The first architecture treats the acceptance and capacity models as feature engineers. Their pre-computed logits, $\eta_{\text{acc}}(i, j) = \mathbf{w}'_{ij}\hat{\boldsymbol{\beta}}$ from the GLMM and $\eta_{\text{cap}}(c, t) = \mathbf{x}'_{ct}\hat{\boldsymbol{\delta}}$ from the capacity model, are passed as the only inputs to a shallow neural network that produces the final acceptance probability:

$$\hat{p}_{\text{final}}^{\text{seq}}(i, j, c, t) = f_{\text{seq}}(\eta_{\text{acc}}(i, j), \eta_{\text{cap}}(c, t)), \quad (11)$$

where $f_{\text{seq}}(\cdot)$ is parameterized as a single hidden layer of eight ReLU units followed by a sigmoid output. The component models' parameters are held fixed during ensemble training; only the second-stage neural network's weights are updated. The architecture is illustrated in Figure 2.

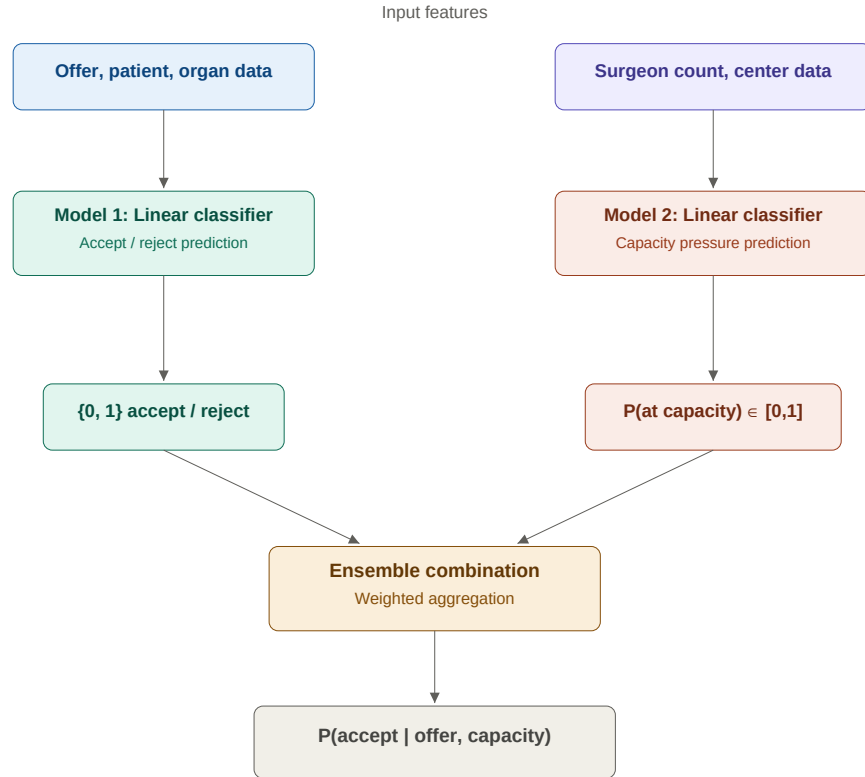


Figure 2: Sequential ensemble architecture. The acceptance model (Model 1) and capacity constraint model (Model 2) are estimated independently and their outputs are passed as inputs to a second-stage neural network that produces the final acceptance probability.

The advantage of this architecture is interpretability: the two component logits are individually meaningful, and the predicted acceptance probability decomposes into a match-driven component and a capacity-driven component that retain the structural meaning given to them by their parent models. The disadvantage is that the components are calibrated for different prediction tasks (offer acceptance for the GLMM, center-month at-capacity classification for the capacity model), so the second-stage network must reconcile signals that were

not jointly optimized for the final outcome.

Single-stage joint logit. The second architecture is an L2-regularized logistic regression that includes both the match-quality features used by the acceptance GLMM and the center-capacity features used by the capacity model, all in a single equation estimated jointly on the offer-level acceptance label:

$$\text{logit}\left(\hat{p}_{\text{final}}^{\text{logit}}(i, j, c, t)\right) = \mathbf{w}'_{ij}\boldsymbol{\beta}^j + \mathbf{x}'_{ct}\boldsymbol{\delta}^j, \quad (12)$$

where $\boldsymbol{\beta}^j$ and $\boldsymbol{\delta}^j$ are coefficient vectors estimated jointly. This specification imposes a constant rate of substitution between match quality and capacity: each feature contributes additively to the log-odds of acceptance, and the marginal effect of one input does not depend on the level of the other. It is the most interpretable of the three specifications, with each feature retaining a single coefficient with a direct log-odds reading, but is also the most restrictive in functional form.

Two-tower neural network. The third architecture is a single-stage neural network with two parallel feature towers and an explicit combination layer. The match tower processes the match-quality features through two hidden layers to produce an embedding $\phi_M(\mathbf{w}_{ij}) \in \mathbb{R}^8$. The capacity tower processes the capacity features through two hidden layers to produce an embedding $\phi_C(\mathbf{x}_{ct}) \in \mathbb{R}^4$. The two embeddings are concatenated and passed through a combination layer that produces the final acceptance probability:

$$\hat{p}_{\text{final}}^{2t}(i, j, c, t) = g([\phi_M(\mathbf{w}_{ij}); \phi_C(\mathbf{x}_{ct})]), \quad (13)$$

where $[\cdot; \cdot]$ denotes concatenation and $g(\cdot)$ is a multilayer perceptron (Linear $\rightarrow$ ReLU $\rightarrow$ Linear $\rightarrow$ Sigmoid) operating on the 12-dimensional concatenated vector. All parameters ϕ_M , ϕ_C , and g are trained end-to-end against the offer-level acceptance label. The architecture is illustrated in

Figure 3.

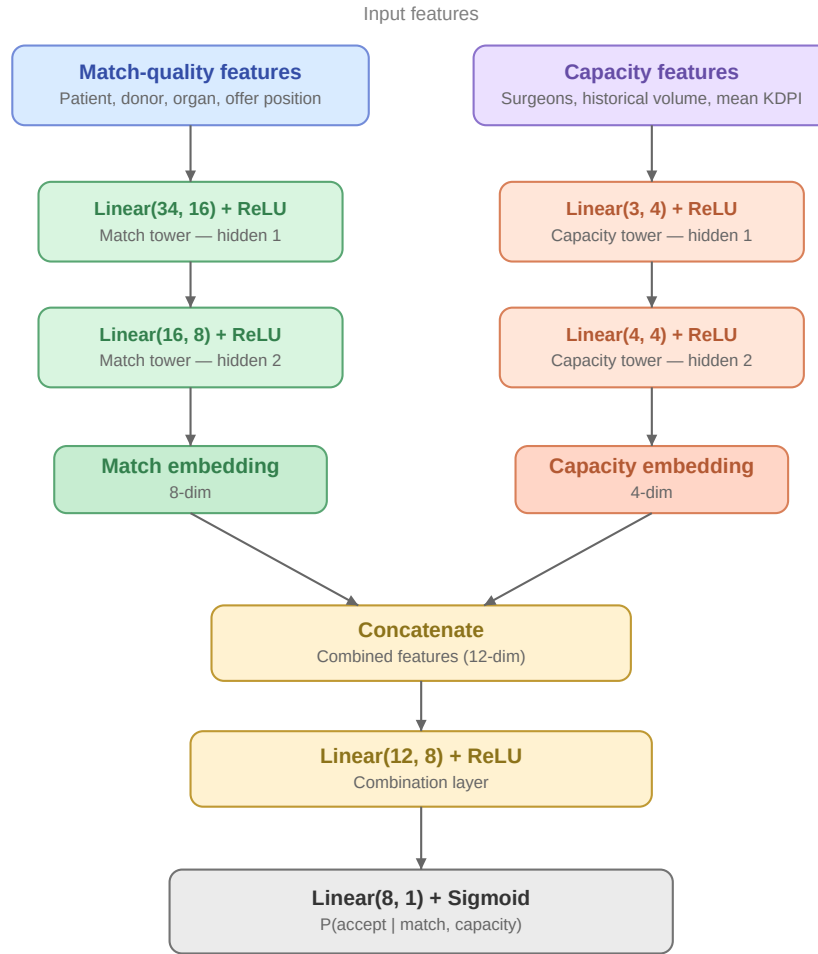


Figure 3: Two-tower neural network architecture. The match-quality features and capacity features are processed by separate towers, each producing a learned embedding. The two embeddings are concatenated and passed through a combination layer that can learn non-linear interactions between the channels, producing the final acceptance probability.

Unlike the joint logit, the two-tower architecture can represent nonlinear interactions between the match and capacity feature groups through the combination layer. This permits acceptance to behave as a nontrivial function of both inputs jointly: the marginal effect of one feature group can depend on the level of the other. The structural intuition the architecture is designed to capture is complementarity—acceptance requires both an attractive match and an available surgeon—which the additive logit cannot encode but which the

two-tower network can learn from data if it is present. The cost is interpretability: the learned embeddings are not directly mappable onto economic constructs, and feature-level coefficients no longer exist.

Comparison and use in simulation. The three architectures are evaluated on the same held-out offers to allow an apples-to-apples comparison of predictive performance. Section 5.9 reports these comparisons in detail. The simulation in Section 4.8 carries forward the architecture whose properties are most useful for the policy-relevant downstream metric: the two-tower network for its predictive accuracy.

4.3.6 Organ-Level Discard Prediction Model

The models described above operate at the offer level: given a specific donor–candidate pair, they predict whether that particular offer will be accepted. A complementary question is whether, at the time a kidney enters the allocation system, we can predict its eventual fate—transplantation or discard—before any offers are made.

We estimate an organ-level discard prediction model that takes as input the characteristics of the donor kidney q_j at the time of recovery and outputs the probability that the kidney will ultimately be discarded:

$$\hat{p}_{\text{disc}}(j) = g(q_j) \in [0, 1]. \quad (14)$$

The input features q_j include donor demographics (age, sex, BMI), clinical measures (KDPI, terminal creatinine, donor comorbidities), procurement characteristics (DCD status, OPO region), and kidney-specific biopsy and perfusion indicators. The training label is the observed binary outcome: whether the kidney was discarded (1) or transplanted (0), as recorded in the STAR donor file.

This model serves two purposes within the broader project. First, it provides a baseline measure of which kidneys are at highest risk of discard under the current system, enabling

targeted evaluation of whether the learned allocation policy disproportionately rescues high-risk organs. Second, it can be embedded in the allocation simulation as an early-warning signal: organs with high predicted discard probability could be routed to alternative, more efficient offer sequences before cold ischemia time accumulates.

4.4 Policy Parameterization

We parameterize the allocation policy using a learnable linear transformation matrix $A \in \mathbb{R}^{d \times p}$. For a candidate i and organ o , define the squared distance

$$d_A(x_i, o)^2 = \|A(x_i - o)\|_2^2. \quad (15)$$

The policy assigns higher priority to candidates closer to the organ under this learned metric. Specifically, the probability of selecting candidate i in state s_t is given by the softmax policy

$$\pi_A(a = i \mid s_t) = \frac{\exp(-d_A(x_i, o_t)^2)}{\sum_{j \in \mathcal{W}_t} \exp(-d_A(x_j, o_t)^2)}. \quad (16)$$

This parameterization preserves interpretability: the learned matrix $\mathbf{A}$ rescales and rotates donor–candidate feature differences, revealing which dimensions of compatibility are emphasized by the optimal policy. The slight stochasticism is meant to account both for the stochasticism in the modeling of acceptance and outcomes, as well as to account for unobserved variables when considering near identical organ recipients.

4.5 Transition Dynamics

The simulation operates in two modes depending on the organ-level discard prediction. When a kidney j enters the system, the discard model produces $\hat{p}_{\text{disc}}(j)$. If $\hat{p}_{\text{disc}}(j)$ falls below a threshold τ —indicating the organ is likely to be transplanted under the status quo—the simulation follows the original match-run ordering from the PTR data and does not invoke

the learned policy. This preserves the baseline allocation for organs that the current system handles adequately.

If $\hat{p}_{\text{disc}}(j) \geq \tau$, the organ is flagged as high-risk for discard and the learned policy π_A is activated. The policy reranks candidates according to the learned distance metric $d_A(x_i, o)^2$, and offers proceed sequentially down the reranked list. At each offer step, acceptance occurs with probability $\hat{p}_{\text{final}}(i, j, c, t)$ as predicted by the ensemble model (or $\hat{p}_{\text{acc}}(i, j)$ from the baseline GLMM, depending on the simulation arm). If the offer is declined, the system transitions to the next candidate on the reranked list. If no candidate accepts within the top- k offers, the organ is discarded.

4.6 Reward Function

The planner’s objective is to maximize the probability that an organ is offered to an accepting candidate before cold ischemia time renders it unusable. A simple terminal reward distinguishing only accepted from discarded outcomes provides no gradient information about *when* during the offer sequence acceptance occurs, leaving the policy indifferent between an early acceptance (low cold ischemia time, high graft quality) and a late acceptance (high cold ischemia time, marginal viability). Because the mechanism that converts a viable kidney into a discard is precisely the accumulation of cold ischemia time across rejected offers, the reward is constructed to reflect that mechanism directly.

Let CIT_t denote the cumulative cold ischemia time after offer t in an episode. The per-step reward is defined as

$$R_t = \mathbb{1}\{\text{offer at step } t \text{ accepted}\} \cdot \exp(-\lambda \cdot \text{CIT}_t) - c \cdot \mathbb{1}\{\text{offer at step } t \text{ rejected}\}, \quad (17)$$

where $\lambda > 0$ controls the rate at which the value of an acceptance decays with cold ischemia time, and $c > 0$ is a small constant penalty applied to each rejection that reflects the cost of consuming time on an offer that did not succeed. The decay parameter λ is calibrated such

that an acceptance after 24 hours of accumulated cold ischemia time yields approximately half the reward of an immediate acceptance.

If the organ is discarded—defined as the cold ischemia time exceeding a viability threshold before any offer is accepted—a terminal penalty is applied:

$$R_{\text{terminal}} = -\beta_D \cdot \mathbb{I}\{\text{discarded}\}. \quad (18)$$

The total return for an episode is the sum of per-step rewards plus the terminal penalty. This formulation builds on prior work using shaped rewards in healthcare RL Naqvi et al. (2025), where reward shaping is used to make sparse outcome signals tractable for policy gradient methods.

This specification gives the policy dense gradient signal at every step rather than only at termination. Early acceptances yield large positive rewards, late acceptances yield smaller but still positive rewards, each rejection along the way contributes a small negative reward, and discard yields a large terminal penalty. The policy therefore learns simultaneously to find acceptors and to find them quickly. Compared with a sparse terminal-only reward, this dense formulation substantially improves learning stability for policy-gradient methods on episodes with high variance in length.

4.7 Learning Algorithm

We optimize the policy parameters $\mathbf{A}$ using the REINFORCE algorithm (Williams 1992), a policy-gradient method, in the simulated environment. Each episode corresponds to an organ arrival, with transitions and rewards generated by the acceptance and survival modules. The matrix $\mathbf{A}$ is initialized to the identity, corresponding to the baseline Euclidean distance, and updated iteratively to maximize expected cumulative reward. Optimization is performed using stochastic gradient ascent with minibatches of simulated organs. This means that the policy is evaluated and updated every n episodes or organ offerings, so we do not update it

due to singular noisy offering incidents.

4.8 Simulation and Evaluation Protocol

To evaluate the learned policy and to quantify the role of surgeon capacity constraints in organ discard, we run the allocation simulation thrice on the same held-out set of organs under three acceptance model and RL allocation specifications:

1. **Capacity-aware arm without RL:** Acceptance decisions are drawn according to $\hat{p}_{\text{final}}(i, j, c, t)$ from the optimal capacity aware model combining the GLMM match criteria with the center capacity pressure estimate $\hat{p}_{\text{cap}}(c, t)$. The RL offering model does not activate for high-discard-risk organs.
2. **Baseline arm (GLMM only) with RL:** Acceptance decisions are drawn according to $\hat{p}_{\text{acc}}(i, j)$ from the Bayesian mixed-effects logistic regression, which does not incorporate center capacity information. The RL offering model activates for high-discard-risk organs.
3. **Capacity-aware arm with RL:** Acceptance decisions are drawn according to $\hat{p}_{\text{final}}(i, j, c, t)$ from the optimal capacity aware model combining the GLMM match criteria with the center capacity pressure estimate $\hat{p}_{\text{cap}}(c, t)$. The RL offering model activates for high-discard-risk organs.

In all arms, the discard prediction gate operates identically: organs with $\hat{p}_{\text{disc}}(j) < \tau$ follow the original match-run order, while organs with $\hat{p}_{\text{disc}}(j) \geq \tau$ are allocated using the learned policy π_A . τ is set to .5. The only differences between arms is the model used to simulate each offer’s acceptance probability and the presence of alternative offering behavior under the RL policy. Comparing discard rates across the two arms isolates the contribution of surgeon capacity constraints: if the capacity-aware arm achieves meaningfully lower discard, it indicates that capacity bottlenecks are a binding constraint that the baseline model fails

to account for. If the two arms produce similar discard rates, it suggests that capacity constraints are either not binding or are already implicitly captured by the center random effect in the GLMM.

During both simulation arms, we maintain a log of every offer instance in which the capacity model predicts that the receiving center is at or near capacity ($\hat{p}_{\text{cap}}(c, t)$ above a specified threshold). This log records the center identifier, time period, urban-rural designation, and the outcome of the offer. We use this log to conduct a post-simulation analysis of capacity constraints by center type. Specifically, we compare urban and rural centers along several dimensions: the frequency of capacity-flagged offers, the average likelihood score of a center being at capacity, the acceptance rate conditional on being flagged, and the average number of offers per unique active surgeon per month. The last measure provides an estimate of workload intensity and helps characterize what constitutes a sustainable offer-to-surgeon ratio across center types.

The primary evaluation metrics are: (i) the discard rate under each arm relative to the observed discard rate in the data, (ii) the number of organs rescued (transplanted under the learned policy that were discarded under the status quo), and (iii) the geographic distribution of capacity-flagged events.

5 Results

This section presents preliminary results from the empirical models described in Section 4.3: the offer-level Bayesian mixed-effects acceptance model and the donor-level discard prediction model, the center capacity model, and the capacity-aware acceptance models. Additionally, the allocation simulation results are analyzed.

5.1 Acceptance Model Estimates

The acceptance model is estimated on a 10% random sample of offers from September–November 2022 (18,929 offers from approximately 1,400 distinct donors), with evaluation on a held-out December 2022 test set (7,189 offers). Tables 1 and 2 report posterior means and standard deviations for the fixed-effects coefficients, categorical indicators, and variance components. All continuous covariates are standardized to zero mean and unit variance, so coefficients represent the change in the log-odds of acceptance associated with a one-standard-deviation increase in the covariate.

Table 1: Acceptance model estimates: fixed effects. Training sample: 18,929 offers (Sep–Nov 2022, 10% sample). Continuous covariates are standardized.

	Post. Mean	Post. SD
Intercept	−2.110	0.058
Patient age at listing	0.271	0.045
Donor age	−0.046	0.051
KDPI	−1.064	0.043
Donor creatinine	−0.530	0.072
Initial cPRA	−0.000	0.041
End cPRA	0.187	0.039
Patient BMI	−0.104	0.077
Years on waitlist	−0.096	0.057
ABO identical match	−0.107	0.046
Log offer position	−0.361	0.041
Log lag offers (4 quarters)	0.014	0.063
Log lag donors (4 quarters)	0.035	0.063
Log center offers (28 days)	−0.273	0.047
Center mean KDPI (28 days)	0.202	0.045

Table 2: Acceptance model estimates: categorical indicators and random effects.

	Post. Mean	Post. SD
<i>Gender^a</i>		
Patient female	−0.876	0.088
Patient male	−1.234	0.076
Donor female	−1.085	0.098
Donor male	−1.025	0.071
<i>Blood type^a</i>		
Patient A / AB / B / O	−0.031 / −0.853 / −0.418 / −0.809	
Donor A / AB / B / O	−1.343 / 0.790 / −1.196 / −0.360	
	SD	95% CI
<i>Random effects</i>		
σ_{donor}	1.329	[1.294, 1.365]
σ_{patient}	0.365	[0.360, 0.370]
σ_{center}	0.830	[0.756, 0.911]

^a Reference category is missing/other.

5.2 Interpretation of Fixed Effects

The largest fixed effect is KDPI (−1.064, SD = 0.043), confirming that organ quality remains the dominant predictor of acceptance: a one-standard-deviation increase in the Kidney Donor Profile Index reduces the log-odds of acceptance by more than one unit. Donor terminal creatinine (−0.530) provides additional negative signal beyond what KDPI captures.

Among patient covariates, age at listing (0.271) is positively associated with acceptance, consistent with the interpretation that older patients face shorter expected remaining time on dialysis and therefore have lower option value from waiting. End cPRA (0.187) is positively

associated with acceptance, while initial cPRA enters with a near-zero coefficient. The end cPRA effect is consistent with the hypothesis that highly sensitized patients receive fewer compatible offers and are therefore less selective when one arrives.

Offer position in the match run (-0.361) enters negatively, as expected. Later positions correspond to candidates who are progressively worse matches, and by the time an organ reaches position 50 or beyond, the remaining candidates are unlikely to accept. The ABO identical match indicator is small and slightly negative in this specification, reflecting the fact that blood type compatibility is largely captured by the patient and donor blood type dummies elsewhere in the model.

5.3 Scarcity and Continuation-Value Features

The local-market scarcity proxies are small in magnitude in this sample: log lagged 4-quarter offer volume (0.014) and unique donor counts (0.035) both have posterior means close to zero with sizeable posterior standard deviations. The expected scarcity-driven selectivity effect is therefore not strongly identified at the OPO–blood-type–wait-time level in the current window.

The center-level features are more informative. The 28-day rolling offer volume for the candidate’s blood type (-0.273) is negatively associated with acceptance, suggesting that centers that have been flooded with recent offers for a given blood type are more selective on the marginal offer. The average KDPI of recent offers at the center (0.202) is positively associated with acceptance: when a center has been receiving lower-quality organs recently, it is more likely to accept the current offer, consistent with a reservation-quality interpretation in which centers calibrate their acceptance threshold to the local supply.

5.4 Random Effects

The variance components reveal substantial unobserved heterogeneity at all three levels, but with a different ordering than the smaller-sample specification reported in earlier drafts. The

donor random effect now has the largest estimated standard deviation ($\sigma_{\text{donor}} = 1.329$, 95% CI [1.294, 1.365]), indicating that organs differ considerably in their underlying desirability beyond what KDPI, creatinine, age, blood type, and DCD status capture. This is the variation attributable to clinical impressions, biopsy findings, and other features communicated informally during the offer process.

The center random effect ($\sigma_{\text{center}} = 0.830$, 95% CI [0.756, 0.911]) remains substantial, indicating that transplant centers differ in their baseline propensity to accept offers even after conditioning on observable organ, patient, and market characteristics. This variation is the target of the capacity model: if surgeon availability and operational constraints can explain a meaningful share of this center-level heterogeneity, the ensemble should produce more accurate and more interpretable acceptance predictions.

The patient random effect ($\sigma_{\text{patient}} = 0.365$) is smaller, reflecting the fact that a single patient typically appears in many offers but with limited variation across them; much of the patient-level heterogeneity is absorbed by observed patient covariates.

5.5 Predictive Performance

Table 3 reports the evaluation of the acceptance model on the held-out Dec. 2022 test set.

Table 3: Predictive performance of the acceptance model on the December 2022 held-out test set (7,189 offers, acceptance rate 1.84%).

Metric	Value
AUC (ROC)	0.857
Log-loss	0.078
Mean predicted probability	0.010
Observed acceptance rate	0.018
Min predicted probability	1.65×10^{-6}
Max predicted probability	0.547

The model achieves an area under the receiver operating characteristic curve (AUC) of 0.857, an improvement over the smaller-sample specification reported in earlier drafts. For a randomly drawn accepted-rejected pair, the model assigns a higher predicted probability to the accepted offer 85.7% of the time. An AUC of this magnitude is strong discriminative performance in a setting with a base acceptance rate below 2%, where the signal-to-noise ratio is inherently low.

The log-loss of 0.078 provides a complementary assessment of calibration. A naive model that predicts the training-set acceptance rate (0.021) for every offer would achieve a log-loss of approximately 0.103. The estimated model improves on this baseline by roughly 25%, a meaningful gain given the difficulty of the prediction task.

The mean predicted probability on the test set (0.010) is somewhat lower than the observed acceptance rate (0.018). As in the earlier specification, this gap arises because the test predictions rely solely on fixed effects; the random intercepts for patients, donors, and centers contribute to in-sample fit but are not carried forward to test-set observations involving previously unseen entities. The discrepancy motivates the ensemble approach: by incorporating center-level capacity information as an observable feature rather than relying on an unobserved random intercept, the capacity model may recover some of the explanatory power that is otherwise lost at prediction time.

5.6 Discard Prediction Model

The discard prediction model is estimated at the donor level on all deceased donors recovered between August and December 2022, restricted to those with at least one kidney recovered ($\text{NUM_KI_RECOV} > 0$). The training sample contains 4,873 donors recovered between August and November 2022, and the test sample contains 1,313 donors recovered in December 2022. The label is constructed as a binary indicator equal to one when zero of the donor's kidneys were ultimately transplanted ($\text{NUM_KI_TX} = 0$ and $\text{NUM_KI_DISC} > 0$). The observed donor-level discard rate of 22.3% is consistent with published OPTN statistics for the pe-

riod (Mohan et al. 2023), providing a sanity check on the label construction. The model is an L2-regularized logistic regression using the same donor-side covariates as the acceptance model: donor age, KDPI, terminal creatinine, hypertension history, DCD status, donor sex, and donor blood type.

Table 4 reports the estimated coefficients.

Table 4: Discard prediction model coefficients. Training sample: 4,873 donors recovered Aug–Nov 2022. Continuous covariates are standardized.

Variable	Coefficient
Intercept	−1.558
Donor age	0.192
KDPI	1.503
Donor creatinine	0.502
Hypertension history	0.158
DCD donor	0.527
Donor female	−0.604
Donor male	−0.828
Donor type A	0.014
Donor type AB	0.502
Donor type B	−0.372
Donor type O	−0.086

The coefficients align with the established clinical literature on organ utilization. KDPI is by far the strongest predictor (1.503): a one-standard-deviation increase in the Kidney Donor Profile Index more than triples the odds of discard. Donor age (0.192) and terminal creatinine (0.502) provide additional positive signal even after conditioning on KDPI, consistent with the observation that KDPI is a useful but imperfect summary of organ quality. The DCD indicator (0.527) shows that donation-after-circulatory-death organs face substantially elevated discard risk, reflecting concerns about warm ischemia damage. Hypertension

history enters with a small positive coefficient, partially absorbed by KDPI which already incorporates donor hypertension. The blood type effects are small, with type AB slightly elevated relative to the missing-category reference and type B slightly depressed; these effects are minor since blood type primarily determines compatibility with recipients rather than organ quality itself.

Table 5 reports the predictive performance of the discard model on the held-out December 2022 test set.

Table 5: Predictive performance of the discard model on the December 2022 held-out test set (1,313 donors, discard rate 25.6%).

Metric	Value
AUC (ROC)	0.847
Log-loss	0.418
Mean predicted probability	0.205
Observed discard rate	0.256
Min predicted probability	0.002
Max predicted probability	0.899

The discard model achieves an AUC of 0.847, indicating that for a randomly drawn discarded–transplanted pair, the model correctly assigns the higher discard probability to the discarded organ 84.7% of the time. As a benchmark, KDPI alone—the existing clinical scoring system designed to summarize discard risk—typically achieves AUC in the range of 0.65–0.75 for predicting kidney non-use. The estimated model therefore improves substantially on the standard clinical benchmark using only a small number of donor-side variables.

The log-loss of 0.418 compares favorably to a naive baseline of approximately 0.520 (predicting the base rate of 0.214 for every donor), an improvement of roughly 20%. The mean predicted probability on the test set (0.205) is close to the observed discard rate (0.256), indicating that the model is reasonably well-calibrated on average. The wide range of test

predictions ($[0.002, 0.899]$) shows that the model is meaningfully separating donors rather than predicting near the base rate uniformly.

For the purposes of the allocation simulation, the discard model serves as a triage gate: organs with high predicted discard probability ($\hat{p}_{\text{disc}}(j) \geq \tau$) are routed through the learned policy, while organs with low predicted discard probability follow the original match-run order. The strong performance of the model on held-out donors supports its use in this role, as it can identify high-risk organs at the time of recovery using only features available before any offers have been made.

5.7 Center-Month Volume Regression

The volume regression replicates the framework of Miller et al. (2024) and extends it with an urban-rural indicator. The specification regresses the monthly count of kidneys transplanted at a center on the natural log of kidneys offered, the number of unique active surgeons, the mean KDRI of offered organs, the mean cold ischemic time (CIT) of transplanted organs, a metropolitan indicator, and two interaction terms: $\log \text{ offered} \times \text{surgeon count}$, and $\log \text{ offered} \times \text{mean KDRI}$. The model is estimated by ordinary least squares on the full panel of center-months from January 2015 through December 2022, providing maximum statistical power to identify the structural relationship between surgeon supply and transplant throughput.

The estimation sample contains 18,732 center-month observations after merging the SRTR offer file, donor characteristics, surgeon supply panel, and metropolitan classifications, and dropping center-months with no offers received. The mean center-month volume is 6.09 kidneys transplanted, with substantial variation across centers and time. Table 6 reports the regression coefficients.

Table 6: OLS regression of center-month transplant volume. Sample: 18,732 center-month observations from January 2015 through December 2022. Standard errors in parentheses.

Variable	Coefficient	Std. Error
Constant	−1.726	0.744
Log kidneys offered	−1.370	0.192
Number of active surgeons	0.331	0.160
Mean KDRI	−3.466	0.502
Mean cold ischemic time	0.292	0.008
Metropolitan center	2.023	0.399
Log offered $\times$ surgeons	0.542	0.035
Log offered $\times$ KDRI	0.940	0.135
R^2		0.511
Adjusted R^2		0.510
F -statistic		2,791

The model explains 51% of the variation in center-month transplant volume. This is below the R^2 of 0.987 reported in the model from Miller et al. (2024), which aggregated to the national-monthly level where idiosyncratic center-level variation is smoothed out. At the center-month level, much of the residual variation can be attributed to center-level effects like scheduling, vacation, and operational fluctuations that are not captured by structural features. An R^2 above 0.5 at this finer granularity nonetheless indicates that the included covariates capture substantial structural signal.

The most important coefficients are the interaction terms, which determine how the marginal effects of offers and quality scale with surgeon supply. The interaction between log offered kidneys and the number of active surgeons is positive and highly significant (0.542, $t = 15.3$), indicating that the value of additional offers increases sharply with surgeon availability. At a center with one active surgeon, the marginal effect of doubling the number of offers is approximately $-1.370 + 0.542 \times 1 = -0.83$ transplants—essentially flat or slightly

negative. At a center with four surgeons the effect is +0.79, and at eight surgeons +2.95. This pattern replicates the central finding of the SRTR poster: surgeon supply binds throughput, and more offers do not translate into more transplants when the surgical workforce is too small to absorb them.

The interaction between log offered kidneys and mean KDRI is also positive and significant (0.940). Smaller centers, which receive fewer offers, are highly sensitive to the average quality of those offers (large negative main effect on KDRI). Larger centers are less affected because their throughput allows them to accept some marginal organs even when average quality is lower. This pattern is consistent with risk tolerance scaling with center capacity.

The metropolitan indicator enters positively and significantly (2.023, $t = 5.1$). Holding surgeon count, offer volume, mean organ quality, and CIT constant, metropolitan centers perform approximately two more transplants per month than non-metropolitan centers. Given a sample mean of 6.09 transplants per month, this represents a roughly one-third advantage for metropolitan centers that operates beyond the channel of surgeon supply alone. This gap is visualized in Figure 4. Likely contributors include access to subspecialty support, established referral networks, and infrastructure that allow metropolitan surgical teams to convert their capacity into completed transplants more efficiently. Axelrod et al. (2008) document substantially lower transplant rates among rural residents and attribute the gap in part to the increasing concentration of transplant services in high-volume urban centers. Sonnenberg et al. (2019) describe the “selective referral” channel through which higher-volume centers attract more referrals, alongside structural and process elements such as multidisciplinary care teams that develop in such settings.

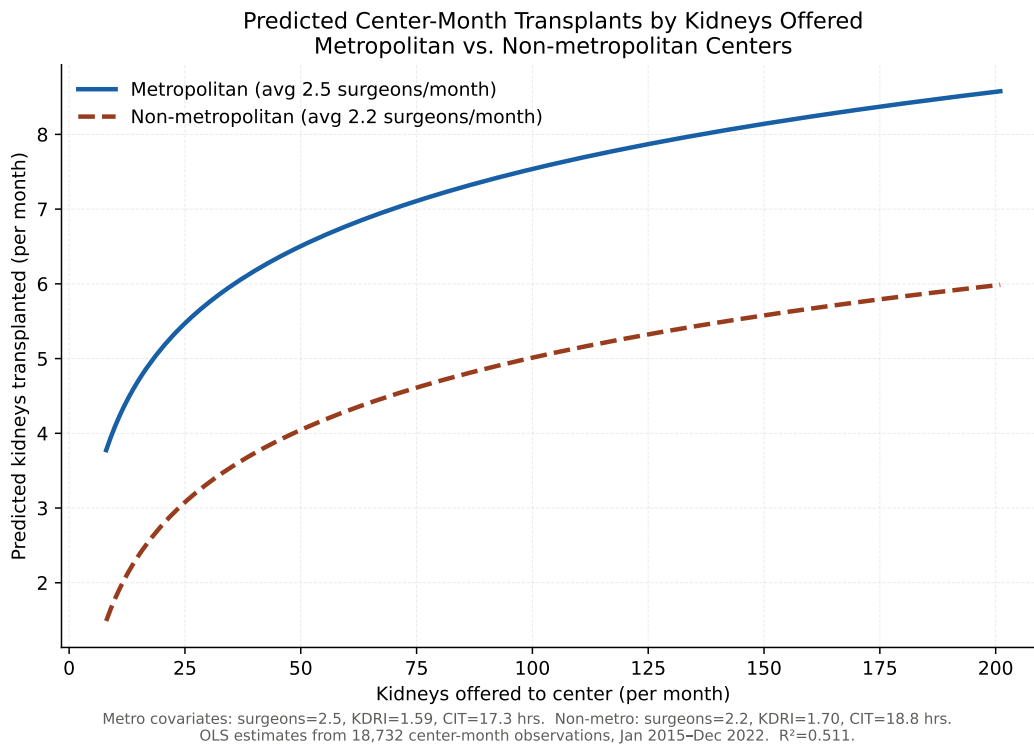


Figure 4: Predicted center-month transplants as a function of offer volume, comparing metropolitan and non-metropolitan centers. Each curve fixes the remaining covariates at the relevant subgroup mean (surgeons, KDRI, CIT). Both curves exhibit the concavity expected of a capacity-constrained throughput function, and metropolitan centers consistently transplant more kidneys than non-metropolitan centers across the full range of offer volumes, even after holding surgeon supply, mean organ quality, and cold ischemia time at the respective groups means.

The mean CIT coefficient (0.292) is positive: high-volume center-months tend to feature transplants drawn from later in the match-run sequence, where organs have traveled farther and accumulated more cold ischemic time. This variable functions here as a model control rather than an actionable lever.

To visualize the central finding, Figure 5 plots predicted transplants per month against kidneys offered for three levels of surgeon supply (the 25th, 50th, and 75th percentiles of the observed surgeon distribution), holding all other covariates at their sample means. The curves are sharply ordered by surgeon count: at any level of offers, centers with more active surgeons convert offers into transplants at substantially higher rates. The gap widens as offers grow, reflecting the positive interaction between offers and surgeons. A center at the

75th percentile of surgeon supply receiving 200 offers in a month is predicted to perform roughly 10 transplants, while a center at the 25th percentile receiving the same number of offers is predicted to perform fewer than 4.

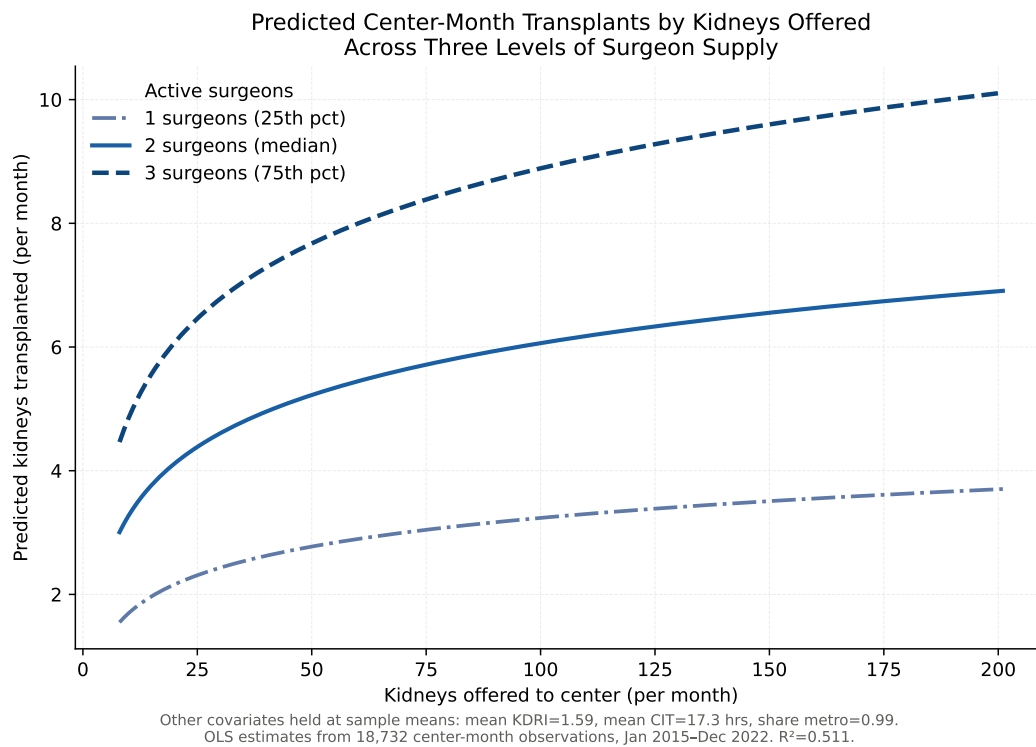


Figure 5: Predicted center-month kidney transplants as a function of monthly kidneys offered, across three levels of surgeon supply. Other covariates are held at sample means. Estimates from OLS regression on 18,732 center-month observations, January 2015 through December 2022.

The volume regression establishes that surgeon supply imposes a meaningful structural ceiling on transplant throughput at the center level. Centers with abundant surgical capacity can absorb additional offers and convert them into transplants; centers with scarce capacity cannot. This finding implies that some share of organ rejections—and therefore some share of organ discards—reflects operational constraints rather than match quality. Identifying which center-months are operating under such constraints is the function of the capacity model developed next.

5.8 Capacity Model

The capacity model classifies each center-month as either operating at or near surgical capacity, or not. Because no direct ground-truth label exists for capacity strain, the label is constructed behaviorally from the offer data. For each center-month cell, the rejection rate is residualized against the log of offers received and the mean KDPI of offers, then a center-month is flagged as at capacity if its residualized rejection rate is in the top quartile within its calendar month. The residualization removes the mechanical relationship between rejection rate and offer volume—centers that receive many offers necessarily reject most of them—and isolates the component of rejection that is unexplained by either offer quantity or offer quality. The within-month quantile cutoff controls for any month-level shifts in baseline rejection patterns. The metropolitan indicator is intentionally excluded from the capacity model so that urban-rural differences can be examined at the simulation stage as an emergent outcome rather than as a baked-in feature.

The estimation sample is restricted to the same window as the acceptance model, August through December 2022, with training on August through November and held-out evaluation on December. After merging the offer panel with the surgeon supply data and the metropolitan classification, and dropping center-months with missing surgeon counts, the analytic sample contains 805 training observations and 202 test observations. The training-sample at-capacity rate is 17.1% and the test-sample rate is 16.3%, both somewhat below the 25% implied by the top-quartile cutoff because of ties at the residualized rejection rate threshold.

The model is an L2-regularized logistic regression with four features: the number of unique active surgeons in the current month, the center’s historical mean monthly transplant volume (computed from PTR data prior to August 2022), the mean offered KDPI in the current month, and the offers received in the current month is excluded as it would re-introduce the leakage that the residualized label was designed to remove. Continuous features are standardized to zero mean and unit variance. Table 7 reports the coefficients.

Table 7: Capacity model coefficients. Sample: 805 center-months from August through November 2022, evaluated on 202 center-months from December 2022. Continuous covariates are standardized.

Variable	Coefficient
Intercept	−3.146
Number of active surgeons	−0.913
Historical monthly volume	−2.617
Mean offered KDPI	0.137

The dominant predictor is historical monthly transplant volume (−2.617): centers that historically perform many transplants per month are far less likely to be flagged as capacity-constrained. Established high-volume centers have built operational infrastructure that allows them to sustain throughput even under heavy offer load. The number of active surgeons is also strongly negative (−0.913): a one-standard-deviation increase in surgeon count cuts the log-odds of being at capacity by nearly one full unit, holding historical volume and offer quality constant. This is the central finding for the capacity channel of the thesis: surgeon supply mechanically reduces the probability of capacity-driven rejection, and does so above and beyond the structural advantage that established high-volume centers enjoy. The small positive coefficient on mean offered KDPI (0.137) indicates that centers receiving lower-quality offers (lower KDPI) reject slightly less often, which is consistent with the residualization control but does not contribute much explanatory power.

Table 8 reports the predictive performance of the capacity model on the held-out December 2022 test set.

Table 8: Predictive performance of the capacity model on the December 2022 held-out test set (202 center-months, at-capacity rate 16.3%).

Metric	Value
AUC (ROC)	0.822
Log-loss	0.345
Mean predicted probability	0.172
Observed at-capacity rate	0.163
Min predicted probability	2.19×10^{-8}
Max predicted probability	0.619

The capacity model achieves an AUC of 0.822 on held-out data, indicating strong discriminative performance. For a randomly drawn at-capacity / not-at-capacity pair of center-months, the model assigns the higher predicted probability to the at-capacity cell 82.2% of the time. The log-loss of 0.345 improves on a naive baseline of approximately 0.456 by about 24%. The mean predicted probability (0.172) is close to the observed at-capacity rate (0.163), indicating reasonable calibration on average. The wide range of predictions across the test set ($[2 \times 10^{-8}, 0.62]$) shows that the model meaningfully separates centers rather than predicting near the base rate uniformly.

A notable structural finding is that the urban-rural divide identified in the volume regression—where metropolitan centers performed roughly two additional transplants per month after full controls—does not manifest as a baked-in feature in the capacity model. This is by design: the metropolitan indicator was deliberately excluded from the capacity model so that urban-rural patterns in capacity strain can emerge organically from the simulation analysis described in Section 4. The combined evidence from the volume regression and the capacity model suggests a coherent two-channel interpretation of the urban-rural gap: metropolitan centers have higher average throughput due to factors beyond surgeon count alone (referral networks, infrastructure, sub-specialty support), while all centers can be relatively strained

against their own resources in a given month. The simulation will allow these two channels to be examined separately by tracking which centers experience capacity events and how those events translate into discard outcomes.

For the purposes of the allocation simulation, the capacity model serves as the second component of the ensemble acceptance predictor. Its output, $\hat{p}_{\text{cap}}(c, t) \in [0, 1]$, is combined with the GLMM acceptance probability $\hat{p}_{\text{acc}}(i, j)$ in the shallow neural network described in Section 4.3 to produce the final acceptance probability used in the simulation arms.

5.9 Combining Match Quality and Capacity: Three Specifications

The acceptance model and the capacity model address related but distinct outcomes. The acceptance model is trained at the offer level on the binary indicator $D_{ij} \in \{0, 1\}$ for whether a particular offer was accepted; its predictions $\hat{p}_{\text{acc}}(i, j)$ capture organ-patient match quality, offer-position dynamics, and local market conditions. The capacity model is trained at the center-month level on a separate, behaviorally-defined indicator for whether a center was operating at or near capacity (the residualized rejection rate cutoff defined in Section 5.8); its predictions $\hat{p}_{\text{cap}}(c, t)$ capture the operational state of the receiving center. Together they cover the two channels through which offer acceptance is hypothesized to operate: organ-patient compatibility and center-level operational availability.

3 methods are fitted and evaluated in order to determine the most empirically sound methodology and economic interpretation of combining these two channels. Three specifications are estimated and compared, each implementing a different position on the spectrum from interpretable structural decomposition to flexible end-to-end learning. All three are evaluated on the same offer subset—the September–November 2022 training window and the December 2022 held-out test window used by the acceptance model, restricted to offers whose (center, month) cell appears in the capacity panel. The analytic sample contains 18,301–18,356 training offers and 7,054–7,084 test offers depending on the specification; small variation is due to the order of dropping operations across pipelines, with 763 offers excluded

in all cases because their center-month lacked surgeon supply data.

Specification 1: Sequential ensemble

The first specification mirrors the architecture introduced in Figure 2: a shallow neural network that consumes the logits from the two pre-trained component models. Specifically, the inputs are $\eta_{\text{acc}}(i, j) = \mathbf{w}'_{ij}\hat{\boldsymbol{\beta}}$ from the acceptance GLMM and $\eta_{\text{cap}}(c, t) = \mathbf{x}'_{ct}\hat{\boldsymbol{\gamma}}$ from the capacity model. These are passed through one hidden layer of eight ReLU units and a sigmoid output. The component models are estimated first and their parameters are frozen; only the neural network's weights are updated during ensemble training, using binary cross-entropy loss, the Adam optimizer at learning rate 0.01, and 50 epochs of mini-batch updates with batch size 512.

This specification treats the component models as feature engineers: each produces a calibrated, interpretable summary of one channel of variation, and the ensemble learns the (potentially nonlinear) mapping from those summaries to the final acceptance probability. The economic interpretation is structural: the predicted acceptance probability decomposes into a match-driven component and a capacity-driven component, each of which retains the meaning given to it by its parent model. Sequential estimation does mean, however, that the two component models are calibrated for different prediction tasks (offer-level acceptance for the GLMM, center-month at-capacity status for the capacity model), and neither has the opportunity to reallocate explanatory power to the other when the ensemble is fit.

Specification 2: Single-stage joint logit

The second specification is an L2-regularized logistic regression that includes both the match-quality features used by the acceptance GLMM and the three center-capacity features used by the capacity model, all in the same equation. All coefficients are estimated jointly on the offer-level acceptance label D_{ij} . This specification imposes a constant rate of substitution between organ quality and capacity: each feature contributes additively to the log-odds of

acceptance, regardless of the values of other features. The marginal effect of an additional unit of KDPI is the same at a center with one surgeon as at a center with eight; the marginal effect of an additional surgeon is the same for a poor match as for an excellent match. Match quality and capacity are treated as substitutes whose contributions add up.

This is the most interpretable of the three specifications: every feature has a single coefficient with a direct log-odds interpretation. It is also the most restrictive in functional form—no nonlinearities, no interactions between feature groups.

Specification 3: Two-tower neural network

The third specification is a single-stage neural network with two parallel feature towers and an explicit combination layer, as detailed in Figure 3. The match tower processes the match-quality features through two hidden layers ($34 \rightarrow 16 \rightarrow 8$ units, ReLU activations) to produce an 8-dimensional embedding. The capacity tower processes the three capacity features through two hidden layers ($3 \rightarrow 4 \rightarrow 4$ units) to produce a 4-dimensional embedding. The two embeddings are concatenated and passed through a combination layer ($12 \rightarrow 8 \rightarrow 1$ units, ReLU then sigmoid). All parameters are trained end-to-end against the offer-level acceptance label using the Adam optimizer with weight decay 10^{-4} , learning rate 10^{-3} , and 80 epochs.

Unlike the single-stage logit, the two-tower architecture can represent nonlinear interactions between the match and capacity feature groups through the combination layer. This permits acceptance to behave as a nontrivial function of both inputs jointly: the marginal effect of one feature group can depend on the level of the other. The structural intuition is complementarity—acceptance requires both an attractive match and an available surgeon—which the additive logit cannot encode but which the two-tower architecture can learn from data if it is present. The cost is interpretability: the learned embeddings are not directly mappable onto economic constructs, and feature-level coefficients no longer exist.

Predictive Performance

Table 9 reports test-set predictive performance for all three specifications, with the GLMM-only baseline included for reference.

Table 9: Predictive performance on the December 2022 held-out test set (approximately 7,054–7,084 offers, acceptance rate 1.86%). All models are evaluated on the same offer subset.^a

Metric	GLMM only	Sequential ensemble	Single-stage logit	Two-tower NN
AUC (ROC)	0.848	0.860	0.879	0.899
Log-loss	0.0790	0.0771	0.0736	0.0684
Mean predicted prob.	0.015	0.021	0.018	0.017
Min predicted prob.	1.6×10^{-6}	2.2×10^{-5}	2.3×10^{-4}	6.5×10^{-5}
Max predicted prob.	0.750	0.428	0.850	0.494

^a The GLMM-only metrics differ slightly from those reported in Table 3 because 763 offers without capacity data are excluded for apples-to-apples comparison with the other specifications.

The four specifications form a strict ordering on both AUC and log-loss. The GLMM alone establishes the baseline at AUC 0.848. The sequential ensemble improves modestly to 0.860, an increase of +0.012. The single-stage joint logit performs substantially better at 0.879, gaining +0.031 over the GLMM and +0.019 over the sequential ensemble. The two-tower neural network achieves the strongest performance at 0.899, gaining +0.051 over the GLMM and +0.039 over the sequential ensemble. Log-loss improves at every step, with the two-tower network achieving a 13% relative reduction over the GLMM-only baseline.

The two-tower neural network is the best-performing model on the metrics reported.

What the Comparison Reveals

The ranking of the three specifications, combined with their structural differences, supports several substantive interpretations.

The gap between the sequential ensemble and the single-stage joint logit (+0.019 AUC) is informative about the cost of two-stage estimation. The two specifications differ along two dimensions: estimation strategy and functional form. The joint logit fits every coefficient on a single offer-level likelihood and combines features additively in log-odds. The sequential ensemble fits its component models in two stages on different labels and then composes them through a hidden layer of eight ReLU units, which permits nonlinear interaction between the two component logits at the combination step. On functional-form grounds alone, then, the sequential ensemble has the more flexible combination layer; the joint logit imposes a strictly additive structure. The fact that the joint logit nonetheless produces meaningfully better predictions therefore cannot be attributed to functional-form flexibility. If anything, that channel runs against the observed result. The remaining explanation is the estimation strategy itself. The capacity model in the sequential ensemble is calibrated for a different prediction task; center-month at-capacity classification rather than offer acceptance. This means the capacity logit it produces is optimized for a related but distinct outcome, and the second-stage network has no opportunity to reallocate explanatory power between the match and capacity channels because the component coefficients are frozen. The joint logit, by contrast, estimates every coefficient against the actual outcome of interest, and the KDPI coefficient in the joint specification is roughly 18% smaller in absolute magnitude than the GLMM's, suggesting that the two-stage approach leaves shared signal between the match and capacity channels unexploited.

The gap between the single-stage joint logit and the two-tower neural network (+0.020 AUC) is informative about functional form. Both specifications are estimated on the same features and the same label in a single stage; the difference is that the joint logit imposes additivity (constant rate of substitution between match quality and capacity) while the two-tower network can learn nonlinear interactions. The fact that the more flexible specification is better is empirical evidence that match quality and capacity do not combine additively: the marginal effect of one depends on the level of the other. The structural reading is that

acceptance is closer to an AND condition than to an additive sum; a high-quality organ-patient match contributes more to the acceptance probability when the receiving center has surgical capacity available, and less when it does not. A constant-rate-of-substitution specification understates the importance of capacity at the high end of match quality and overstates it at the low end. The two-tower network can represent this complementarity; the additive logit cannot.

A separate diagnostic supports the same conclusion. The maximum predicted probability of the GLMM-only baseline is 0.750, and the joint logit's maximum is 0.850; both specifications produce confident predictions for organ-patient pairs whose match characteristics suggest high acceptance probability. The sequential ensemble's maximum drops to 0.428 and the two-tower network's to 0.494. The compression at the high end is the signature of a model that has learned moderation between match quality and capacity: confident match-driven predictions are tempered when capacity pressure is also high. The sequential ensemble (through the structure of the second-stage neural network) and the two-tower network (through the combination layer) models exhibit this compression, which can be attributed to both incorporating some form of nonlinear interaction. Conversely, the GLMM and the joint logit both impose additivity and therefore cannot represent moderation between channels.

Among the three specifications, the two-tower network is the most accurate predictor and provides the most direct evidence for complementarity between match quality and capacity. The single-stage joint logit is the most interpretable of the high-performing specifications, with each feature retaining a clean log-odds coefficient, and provides the cleanest contrast for the functional-form comparison.

For this reason, the two-tower network is utilized in the simulation analysis described in Section 4. The two-tower network supports the highest-fidelity predictions of which offers are likely to be accepted. The standalone capacity model will be used to track how often specific centers are likely near capacity.

5.10 Reinforcement Learning Policy

The policy parameters $\mathbf{A} \in \mathbb{R}^{7 \times 7}$ are trained on 12,749 kidney arrivals from January through November 2022, with 456 kidneys recovered in December 2022 held out for evaluation. Training proceeds for five epochs of REINFORCE updates with batch size 64, learning rate 0.01, and gradient clipping at norm 1.0. The policy is initialized at $A = I$, corresponding to candidate ranking by standard Euclidean distance over the seven-dimensional feature space.

Training behavior

Mean episode return remains in a narrow band of 0.61 to 0.67 across all five training epochs, with no clear monotonic trend. Acceptance rates across training are similarly stable at 92% to 95%. The narrow range reflects two features of the simulation environment. First, the eligible candidate pool for any given organ is large, often spanning hundreds or thousands of medically compatible patients, so finding at least one accepting candidate within the maximum offer cap is nearly always feasible regardless of policy structure. Second, the identity-matrix initialization is already a sensible baseline because all features have been standardized, so Euclidean distance is a reasonable proxy for similarity from the outset. The policy gradient signal therefore operates on the margin of *which* candidates the policy offers first within those successful episodes rather than *whether* the episode is successful at all.

Test-set performance

Evaluated on the 456 December 2022 test organs, the trained policy achieves an acceptance rate of 92.3% (corresponding to a simulated discard rate of 7.7%), with a mean of 14.6 offers per episode and a mean cumulative cold ischemia time of 6.84 hours at termination. These figures should not be compared directly to real-world discard rates, which are influenced by operational frictions absent from the simulation; their interpretation is reserved for the head-to-head comparison across simulation arms in Section 5.11.

What the policy learned

Although training reward is flat, the policy nonetheless reorganizes the distance metric substantially. The trained transformation matrix $\mathbf{A}$ differs from its identity initialization by a relative Frobenius norm of $\|\mathbf{A} - \mathbf{I}\|_F / \|\mathbf{I}\|_F = 0.92$. This suggests that the gradient signal, while small in magnitude relative to sampling noise on the reward, carried consistent information about which feature directions to emphasize in the distance metric.

Figure 6 presents the trained matrix as a heat map. Diagonal entries remain dominantly positive, indicating that the policy retains the basic structure of Euclidean distance over each input feature, but off-diagonal entries reveal substantial cross-feature mixing.

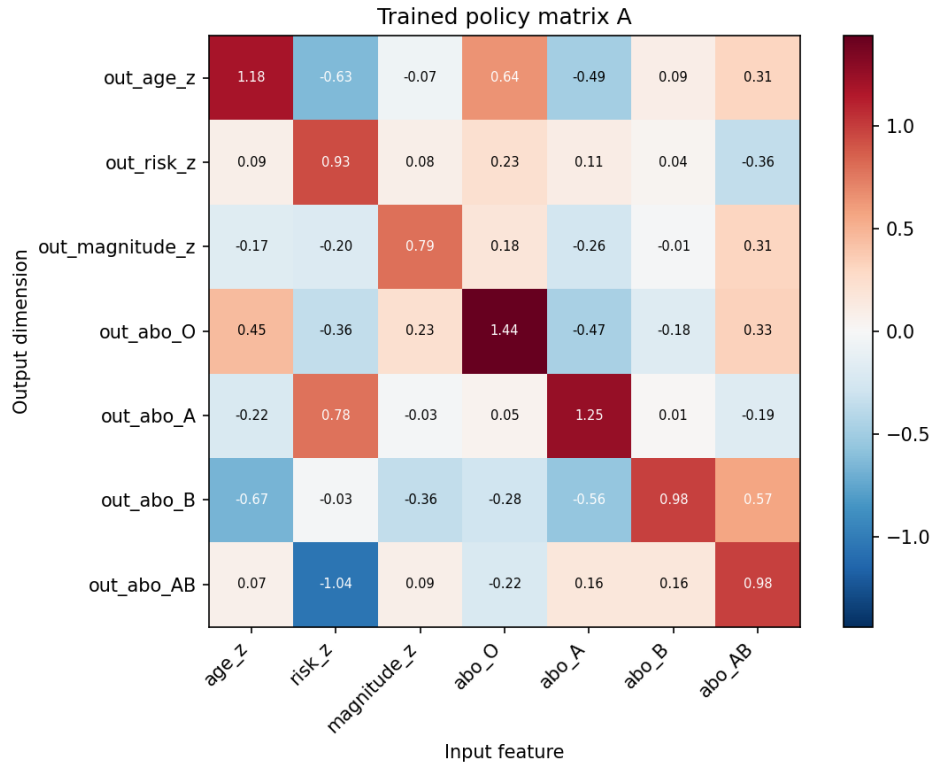


Figure 6: Trained policy transformation matrix $\mathbf{A}$. Rows correspond to output dimensions of the projected feature space; columns correspond to input feature differences between candidate and organ. Cells show the learned weight by which each input feature contributes to each output dimension. Cells colored deep red indicate strong positive loadings, deep blue strong negative loadings. The diagonal remains dominantly positive but substantial off-diagonal entries reveal cross-feature mixing learned during training.

The per-feature column norms of A , obtained by summing squared column entries, indicate how strongly each input feature contributes to the learned distance metric. Identity initialization assigns each feature a column norm of exactly 1.0. The trained policy upweights several features substantially: the risk-axis feature (CPRA for patients, KDPI for donors) has column norm 3.13, more than triple its initial weight; the O blood-type indicator has column norm 2.69; the A blood-type indicator 2.44; and patient/donor age 2.14. The AB blood-type indicator increases modestly to 1.74, while the B blood-type indicator remains essentially at its initial weight at 1.02, and the magnitude-axis feature (BMI for patients, terminal creatinine for donors) decreases slightly to 0.82.

The interpretation is that the policy learned to weight risk-axis matching most heavily within the medically compatible pool, with secondary emphasis on finer-grained ABO sub-categorization and age-matching, and effectively to discount the body-mass and creatinine dimension as less informative for predicting acceptance. The asymmetric ABO weighting also has a sensible structural interpretation: O is the universal donor blood type and A is the most common type after O, so finer matching within these categories has the greatest scope to differentiate among many compatible candidates, whereas B is the rarest type and therefore offers fewer candidates for the metric to discriminate among.

A singular value decomposition of A provides a complementary view of the learned structure. All seven singular values exceed 5% of the largest, indicating that the policy uses the full dimensionality of the feature space rather than collapsing onto a low-rank subspace. The largest singular value is 2.51 and corresponds to a right singular vector that loads positively on risk (+0.61) and the A blood-type indicator (+0.45), and negatively on age and the O and AB blood-type indicators. This dominant direction in the learned distance metric simultaneously distinguishes high-risk from low-risk candidates and A-typed from non-A-typed candidates, suggesting the policy identified an interaction between organ risk and ABO sub-category as the most important axis of compatibility within the medically compatible pool.

5.11 Allocation Simulation

To evaluate the contribution of the learned policy and the choice of acceptance model on allocation outcomes, the trained policy and the empirical models are embedded in a counterfactual simulation environment that replays the December 2022 allocation period under three regimes. The simulation operates on the 456 donor kidneys recovered in December 2022 and traces, for each kidney, the sequence of offers that would be made under each regime, the acceptance outcome of each offer, and the final disposition of the organ.

Real-world benchmark

The actual observed disposition of kidneys recovered in December 2022 provides the empirical reference point against which simulation results are interpreted. Drawing on the STAR donor file for the same window, 1,313 donors had at least one kidney recovered in December 2022, yielding 2,612 recovered kidneys in total. Of these, 1,808 were transplanted and 804 were discarded, corresponding to an observed discard rate of 30.78%. This is somewhat higher than the long-run national average of approximately 20% to 22%, consistent with documented seasonal effects in late-year transplant volume associated with holiday-period staffing constraints and reduced operating room availability.

The 30.78% figure is reported to motivate the policy question rather than as a target the simulation seeks to reproduce. Three considerations argue against direct comparison between simulated and observed discard rates. First, the simulation operates on a more permissive eligible candidate pool, defined by blood-type compatibility within the active waitlist at the donor’s recovery date and capped at 2,000 candidates per organ. The actual allocation algorithm applies additional hard constraints, including HLA antigen matching, geographic proximity, and time-based prioritization, that are not modeled here. Second, the simulator uses a deterministic cold ischemia time accumulation model in which each offer adds a fixed number of hours, abstracting away from the heterogeneous logistical delays that drive a substantial share of real-world discards. Third, the acceptance models, while

calibrated against historical data, were trained on offers that did occur and may overstate acceptance probabilities for candidates who in practice would never have received an offer. Together these simplifications imply that the simulation occupies a more favorable corner of the allocation environment than reality, and absolute discard rates produced by any simulation arm are expected to be lower than the 30.78% observed.

The simulation is therefore most informative through its *between-arm* comparisons rather than its absolute outputs. The three arms differ only in the acceptance model used and in whether the trained policy is invoked, with all other elements of the simulation environment held fixed. Differences across arms isolate the marginal contribution of the capacity-aware acceptance model and of the learned reranking policy, conditional on a common simulation environment.

Methodological note: GLMM intercept calibration

A technical adjustment is applied to the GLMM at simulation time. The acceptance GLMM is a mixed-effects model with random intercepts for patient, donor, and center. Training-time predictions on the held-out test set incorporate these random intercepts and produce a mean predicted acceptance probability of 0.0186, matching the empirical offer-level acceptance rate. At simulation time, however, predictions cannot include random intercepts because the relevant patient, donor, and center triples often differ from those observed at training. Setting the random effects to zero and using only the fixed-effects component of the linear predictor produces what Zeger et al. (1988) term population-averaged rather than subject-specific predictions, and the conditional fixed-effects coefficients of a random-intercept model are systematically larger in absolute value than the corresponding marginal coefficients by an attenuation factor of $1/\sqrt{c^2\sigma_u^2 + 1}$, where σ_u^2 is the variance of the random effects and $c = 16\sqrt{3}/(15\pi)$. Pavlou et al. (2015) demonstrate via simulation that using conditional fixed-effects coefficients without correction to produce population-averaged predictions yields predictions that are miscalibrated in the mean (calibration in the large $\neq 0$) and excessively

dispersed (calibration slope < 1), with the magnitude of miscalibration growing as the intra-cluster correlation increases.

To partially correct for this, the GLMM intercept is shifted by an additive constant chosen so that the mean predicted acceptance probability over a representative sample of December 2022 offers equals the training-time mean of 0.0186. The shift addresses the calibration-in-the-large component of the miscalibration described above, restoring the mean of predicted probabilities to its training-time level while preserving the GLMM’s relative ranking of candidates and therefore its discriminative ability. A more complete correction following the rescaling approach validated by Pavlou et al. (2015) would also adjust the slope coefficients, but this is unnecessary in the present application because the simulation evaluation depends primarily on the relative ranking of candidates rather than on the absolute values of predicted probabilities. The two-tower model requires no analogous adjustment because it has no random intercepts.

Simulation arms

The three simulation arms differ along two dimensions: the acceptance model used to score each offer, and whether the learned policy is permitted to rerank candidates. The arms are:

1. **Arm 1 (Two-Tower, no policy):** For each test organ, offers proceed in the order recorded in the SRTR Potential Transplant Recipient (PTR) file. Each offer’s acceptance probability is produced by the two-tower neural network. This arm establishes the simulated baseline of what would occur under the existing match-run order if the two-tower model determined acceptance outcomes.
2. **Arm 2 (GLMM with policy):** The discard model is queried at the start of each episode. If the predicted discard probability exceeds the threshold $\tau = 0.5$, the learned policy is activated and reranks the eligible candidate pool from the KIDPAN waitlist. Otherwise, the original PTR order is followed. Each offer’s acceptance probability is produced by the calibrated GLMM.

3. **Arm 3 (Two-Tower with policy):** Identical to Arm 2 in policy gating, but each offer’s acceptance probability is produced by the two-tower model. This arm tests the joint contribution of capacity-aware acceptance modeling and learned policy reranking.

The $\tau = 0.5$ threshold is chosen to limit policy activation to organs at substantial risk of discard, preserving the existing allocation order for organs that the system handles well under the status quo. At this threshold, the policy activates on 103 of 456 test organs, or 22.6% of the simulation sample.

Three-arm comparison

Table 10 reports the three arms’ aggregate outcomes on the December 2022 test set.

Table 10: Simulation outcomes by arm. All arms are evaluated on the same 456 December 2022 test organs.

Outcome	Arm 1 (TT, no RL)	Arm 2 (GLMM + RL)	Arm 3 (TT + RL)
Discard rate	0.193	0.472	0.180
Mean offers per organ	9.28	27.44	10.32
Mean cold ischemia time (hrs)	4.24	13.46	4.75
RL activation rate	0.0%	22.6%	22.6%
Episode count	456	456	456

Two findings are visible in the table.

The learned policy reduces discards under the two-tower model. Arm 3 attains a simulated discard rate of 0.180, against 0.193 for Arm 1, a reduction of 1.3 percentage points. With 456 test organs, the standard error on the discard-rate estimate is approximately $\sqrt{0.18 \times 0.82/456} \approx 0.018$, so the gap is on the order of one standard error and should be interpreted as suggestive rather than definitive at this sample size. The direction is nonetheless consistent with the policy’s intended function. The policy activates on 103

organs (22.6% of the sample), corresponding to those with predicted discard probability exceeding $\tau = 0.5$. Translating the rate difference into rescued organs, the policy converts approximately $0.013 \times 456 \approx 6$ additional kidneys from discard to transplant in the simulated December 2022 window.

Capacity-aware acceptance modeling is essential. Arm 2, which uses the GLMM as the acceptance model in place of the two-tower, attains a simulated discard rate of 0.472, more than 2.6 times the rate observed under either two-tower arm. The mean number of offers required per organ in Arm 2 is 27.4, against 9.3 for Arm 1 and 10.3 for Arm 3. The mean cold ischemia time at termination is correspondingly higher in Arm 2 (13.5 hours) than in Arms 1 or 3 (approximately 4.5 hours). These results indicate that, although the GLMM achieves a competitive AUC of 0.857 on individual-offer prediction, its rank-ordering of candidates is sufficiently less accurate than the two-tower’s 0.899 to substantially degrade allocation outcomes when used to drive a sequential allocation simulation. The compounding effect of small per-offer ranking errors over the 30-or-more-offer typical episode produces a meaningfully larger discard rate.

The interpretation of this gap is that capacity-aware modeling matters not only for the marginal predictive accuracy on individual offers but for the aggregate efficiency of the allocation system. Even a relatively small improvement in offer-level discrimination, when applied at scale across thousands of sequential decisions per organ, accumulates into a substantial difference in policy-relevant outcomes.

Decomposing the contributions. Comparing arms pairwise isolates the marginal contributions of the two design choices.

- Arm 3 versus Arm 1 holds the acceptance model fixed at the two-tower and varies whether the policy is active. The difference of -1.3 percentage points in discard rate isolates the marginal contribution of the learned policy.

- Arm 3 versus Arm 2 holds the policy fixed (active under the same gating rule) and varies the acceptance model. The difference of -29.2 percentage points isolates the marginal contribution of capacity-aware acceptance modeling.

The latter contribution is roughly twenty times larger than the former in absolute magnitude. While the policy contribution is in the right direction and meaningful as a proportional reduction in discards (a 7% relative reduction over Arm 1), the dominant simulation-relevant signal is the choice of acceptance model. The two-tower model’s incorporation of center-level capacity features confers substantially more value than reranking offers under a non-capacity-aware acceptance model.

Comparison to real-world benchmark. Both two-tower arms produce simulated discard rates approximately 11 to 13 percentage points below the real-world December 2022 rate of 30.78%. As discussed in the methodology, this gap reflects simplifications in the simulation environment rather than the model’s standalone performance, and absolute simulated discard rates should not be interpreted as predictions for real-world deployment. The Arm 2 simulated rate of 47% exceeds the observed rate, but this is consistent with a simulation in which an acceptance model lacking capacity information is asked to drive sequential allocation decisions over the same generous candidate pool: the per-offer probabilities are too compressed for the cumulative acceptance probability to reach realistic levels even over many offers. The gap between Arm 2 and Arm 3 is therefore best read as a model-comparison statement about the value added by the capacity-aware modeling layer, conditional on the simulation environment, rather than as an estimate of system performance.

5.12 Capacity-Acceptance Coupling at the Offer Level

A core architectural claim of the two-tower acceptance model is that combining a match tower with a capacity tower produces predictions that respond to capacity pressure as a depressing factor on acceptance probability. To check whether this claim holds in practice

on the simulation outputs, the offer-level correlation between simulated capacity probability $\hat{p}_{\text{cap}}$ and predicted acceptance probability $\hat{p}_{\text{accept}}$ is computed across all 4,152 Arm 3 offers with both quantities defined. Table 11 reports the Pearson and Spearman correlations, both overall and stratified by whether the RL policy was active for the episode.

Table 11: Offer-level correlation between $\hat{p}_{\text{cap}}$ and $\hat{p}_{\text{accept}}$ in Arm 3.

Subset	n_{offers}	Pearson r	Spearman ρ
All offers	4,152	−0.359	−0.449
RL inactive (rl_act = 0)	2,629	−0.456	—
RL active (rl_act = 1)	1,523	−0.374	—

The negative correlation is sizable in magnitude and consistent across routing modes, indicating that the two-tower model assigns systematically lower acceptance probabilities to offers made under high simulated capacity pressure. The Spearman rank correlation is somewhat stronger than the Pearson coefficient, reflecting the long right tail of the $\hat{p}_{\text{cap}}$ distribution. The fact that the correlation magnitudes are similar in both RL-inactive and RL-active subsets confirms that the capacity-acceptance coupling is a property of the acceptance model itself rather than an artifact of how the RL policy orders candidates.

Table 12 stratifies the comparison further by quartile of $\hat{p}_{\text{cap}}$. Mean predicted acceptance probability falls monotonically across the four quartiles, from 12.7% in the lowest-pressure quartile to 4.7% in the highest, a 2.7-fold reduction. This pattern is consistent with the model architecture: when the capacity tower’s signal indicates a high-pressure center-month, the combined network reduces the candidate’s acceptance probability accordingly.

Table 12: Mean $\hat{p}_{\text{accept}}$ by quartile of $\hat{p}_{\text{cap}}$, Arm 3.

$\hat{p}_{\text{cap}}$ quartile	n_{offers}	Mean $\hat{p}_{\text{cap}}$	Mean $\hat{p}_{\text{accept}}$
Q1 (lowest)	1,129	0.001	0.127
Q2	958	0.008	0.094
Q3	1,027	0.061	0.064
Q4 (highest)	1,038	0.333	0.047

This offer-level coupling provides the mechanistic basis for the system-level patterns reported in Section 5.13: the two-tower model is identifying high-capacity-pressure centers and depressing the acceptance probability of offers routed there, which in turn allows the simulation to direct organs toward lower-pressure centers where acceptance is more likely. The capacity-aware acceptance model is functioning as designed.

5.13 Geographic Analysis: Center-Level and State-Level

This final section examines whether the simulation outputs reveal systematic differences in capacity pressure or organ discard along the urban-rural dimension. Two complementary analyses are presented: a center-level analysis using the SRTR-provided metropolitan classification, and a state-level analysis using U.S. Census 2020 rural-population shares. The center-level analysis is presented first, followed by a discussion of its data limitations, which motivate the broader state-level treatment.

Center-Level Metropolitan Analysis. Each transplant center in the SRTR data is classified as either metropolitan or non-metropolitan based on the geographic location of the center (Section 3.5). Aggregating the Arm 3 simulation outputs by this classification yields the comparison shown in Table 13. Mean simulated capacity pressure at non-metropolitan centers is more than three times that at metropolitan centers, while the organ-level discard rate among organs that visited at least one non-metropolitan center is substantially lower

than the rate among organs offered exclusively to metropolitan centers. The same directional pattern appears across all three simulation arms (Table 14): non-metropolitan centers consistently exhibit two-to-four-fold higher mean simulated capacity pressure than metropolitan centers, regardless of acceptance model or policy regime.

Table 13: Center-level metropolitan comparison, Arm 3.

	n_{offers}	n_{centers}	Mean $\hat{p}_{\text{cap}}$	Discard Rate
Metropolitan	4,134	190	0.099	13.5%
Non-metropolitan	18	3	0.346	5.0%
Ratio (non-metro / metro)	—	—	$3.5\times$	$0.37\times$

Table 14: Mean simulated capacity pressure ($\hat{p}_{\text{cap}}$) at metropolitan vs. non-metropolitan centers, all arms.

	Metro	Non-metro	Ratio
Arm 1 (Two-tower, no RL)	0.109	0.233	$2.1\times$
Arm 2 (GLMM + RL)	0.086	0.340	$4.0\times$
Arm 3 (Two-tower + RL)	0.099	0.346	$3.5\times$

Limitations of the Center-Level Analysis. Despite this consistent directional signal, the center-level analysis is severely limited by the structure of the U.S. transplant system. Of the 254 transplant centers represented in the SRTR data, only three are classified as non-metropolitan: NHDH (Dartmouth-Hitchcock, New Hampshire), SCLA (South Carolina), and TXJS (Texas). Across the December 2022 simulation period, the Arm 3 logs contain just 36 offers to non-metropolitan centers covering 20 organs that visited at least one such center. With this many centers and offers, the simulated capacity-pressure and discard-rate comparisons cannot support inferential claims about the population of non-metropolitan transplant centers; they describe the behavior of three specific centers in one calendar month.

The classification is also coarse: a metropolitan center in a small Appalachian city is grouped with one in central Manhattan despite plausibly different surgeon-supply and patient-volume environments. To obtain a higher-powered geographic comparison that captures gradient variation in rurality, the analysis is extended to the state level.

State-Level Rurality Measure. State rurality is measured as the percentage of state population residing in rural areas in the 2020 Decennial Census, where rural is defined as the population residing outside of areas classified as urban under the post-2020 Census criteria (U.S. Census Bureau 2022). Sums of within-state urban populations across the 50 states and the District of Columbia reconcile to the Census Bureau’s published national total of 265,149,027. State of each offer is identified from the first two letters of the four-letter center code, which encode the standard two-letter U.S. postal abbreviation. After restricting to offers with both a valid simulated capacity probability and a known metropolitan classification, the Arm 3 logs cover 47 of the 51 jurisdictions, with offer counts ranging from 2 (North Dakota, West Virginia) to 411 (New York). Vermont (64.9% rural) is the most rural state represented; California (5.8% rural) is the least.

State-Level Correlations. Pearson correlations between state rurality and the two simulation outcomes are reported in Table 15. The capacity-pressure correlation appears strong in the full sample but collapses under sample-size restrictions, indicating that it is driven by low-volume rural states whose state means are determined by a single center. The discard-rate correlation runs in the opposite direction and *strengthens* after excluding low-volume states, suggesting a more robust relationship between higher state rurality and lower simulated discard.

Table 15: Pearson correlations between state rurality and simulation outcomes (Arm 3), under sample-size restrictions.

Sample restriction	n_{states}	Rural % vs $\hat{p}_{\text{cap}}$	Rural % vs Discard
All represented states	47	+0.41	−0.12
States with ≥ 10 offers	38	+0.11	−0.43
States with ≥ 25 offers	30	+0.13	−0.36
States with ≥ 50 offers	22	+0.04	−0.43

Comparing High-Rurality and Low-Rurality States. To make the discard-rate finding concrete, Table 16 contrasts the most rural and least rural states with substantial offer volume in the Arm 3 simulation. Rural states with $n \geq 50$ offers have a mean discard rate of approximately 4%, while urban states with $n \geq 100$ offers cluster much higher, with several exceeding 20%. This contrast is consistent with the center-level finding: the capacity-aware model in Arm 3 routes offers to rural-state centers efficiently, while organs offered through long chains at high-volume urban centers experience compounding rejection probability.

Table 16: Discard rates in Arm 3 for high-volume rural and urban states.

State	% Rural	n_{offers}	Discard Rate
<i>Most rural states with substantial offer volume</i>			
Mississippi	53.7	21	0.0%
Iowa	36.9	38	2.6%
Tennessee	33.8	71	5.6%
South Carolina	32.1	165	4.9%
Missouri	30.5	78	14.1%
<i>Most urban states with substantial offer volume</i>			
California	5.8	396	9.6%
New Jersey	6.3	117	16.2%
Florida	8.5	147	12.9%
Massachusetts	8.7	123	46.3%
New York	12.6	411	20.7%

Interpretation. One reading of these patterns is that the capacity-aware two-tower acceptance model, by predicting which centers are at high simulated capacity pressure, successfully steers offers away from overloaded centers and toward those with sufficient slack to accept. Rural-state centers, despite operating with lower surgeon counts on average, are not the bottleneck in the simulated allocation: the model identifies which rural-state candidates are good fits and accepts there efficiently, while organs offered to high-volume metropolitan centers experience substantial competition from many candidates and the simulation’s binary accept/reject draws compound across longer offer chains. The state-level pattern is consistent with the center-level finding above: organs visiting non-metropolitan centers in Arm 3 are discarded at a lower rate than organs offered only to metropolitan centers (Table 13). The state-level capacity-pressure correlation is reported here primarily as a robustness check that did *not* confirm a hypothesized relationship, and the negative discard correlation is re-

ported as a substantive observation about the behavior of the capacity-aware model on the December 2022 simulation period. Neither finding establishes a causal mechanism, and both should be interpreted in the context of the simulation’s reliance on a single calendar month of offer data and a fixed set of acceptance and capacity models trained on prior periods.

6 Conclusion

This paper investigates the U.S. deceased-donor kidney allocation system through the lens of capacity-constrained acceptance modeling and learned reranking policies. The central question is whether incorporating center-level surgical capacity into offer-level acceptance prediction, combined with reinforcement-learning-based reranking for high-discard-risk organs, can reduce the simulated organ discard rate. Using SRTR data on offers, donors, and patients, paired with a panel of monthly surgeon counts and metropolitan classifications for U.S. transplant centers, the analysis estimates a sequence of empirical models—a Bayesian mixed-effects acceptance GLMM, a donor-level discard prediction model, a center-month capacity model, three specifications combining match quality and capacity, and a REINFORCE-trained policy over a learned distance metric—and embeds them in a counterfactual simulation of the December 2022 allocation period.

6.1 Summary of Findings

Surgeon supply imposes a structural ceiling on transplant volume. The center-month volume regression replicates and extends the framework of Miller et al. (2024) on a panel of 18,732 center-months from January 2015 through December 2022. The interaction between log offered kidneys and surgeon count is large and highly significant ($+0.542$, $t = 15.3$): the marginal effect of additional offers on transplant throughput is essentially zero at single-surgeon centers but becomes substantial at centers with four or more active surgeons. This finding establishes that the surgical workforce is a binding capacity constraint at the

center level and that some share of organ rejections must reflect operational rather than match-quality considerations.

The metropolitan-rural gap in throughput is not fully explained by surgeon count. The volume regression also identifies a metropolitan effect of +2.023 transplants per month ($t = 5.1$) that persists after conditioning on surgeon supply, offer volume, mean offered KDRI, and cold ischemia time. With a sample-mean throughput of 6.09 transplants per month, this represents a roughly one-third advantage for metropolitan centers operating beyond the channel of surgeon count alone. Plausible contributors include access to subspecialty support, established referral networks, and clinical infrastructure documented by Axelrod et al. (2008) and Sonnenberg et al. (2019), but the precise mechanism is not identified by the present analysis.

Match quality and surgical capacity behave as complements, not substitutes. Three specifications combining match quality and capacity were estimated and compared on a common offer subset: a sequential ensemble that consumes the logits of the pre-trained acceptance and capacity models, a single-stage joint logit imposing additivity, and a two-tower neural network that learns nonlinear interactions between match and capacity feature groups. The two-tower architecture achieves the highest test-set AUC (0.899) and the lowest log-loss (0.0684), improving on the GLMM-only baseline by +0.051 AUC and on the sequential ensemble by +0.039 AUC. The economic interpretation of this ordering is structural: the joint logit imposes a constant rate of substitution between match quality and capacity (each feature contributes additively to the log-odds of acceptance regardless of the values of other features), whereas the two-tower combination layer can represent nonlinear interactions across feature groups. The fact that the more flexible specification meaningfully outperforms the additive one is empirical evidence that match quality and capacity do not trade off at a constant rate; they combine as complementary inputs into the acceptance decision. A strong organ-patient match contributes more to acceptance probability when the receiving center

has surgical capacity available, and less when it does not, which is consistent with how acceptance is actually produced at transplant centers: a good match alone cannot be transplanted without an available surgeon, and surgical availability alone cannot generate an acceptance without a clinically appropriate match. The constant-rate-of-substitution specification understates the importance of capacity at the high end of match quality and overstates it at the low end, which is the source of the AUC gap. A complementary diagnostic supports the same interpretation: the maximum predicted probability under the two-tower model is 0.494, against 0.850 for the joint logit and 0.750 for the GLMM. The compression at the high end is the signature of a model that has learned to moderate confident match-driven predictions when capacity pressure is also high, which the additive specifications cannot do.

Capacity-aware modeling matters more than reranking under the simulation.

The three-arm simulation isolates the marginal contribution of each design choice. Comparing Arm 3 (two-tower acceptance with the learned policy) to Arm 1 (two-tower with the original PTR order) yields a discard-rate reduction of 1.3 percentage points attributable to the policy itself. Comparing Arm 3 to Arm 2 (GLMM acceptance with the same policy gating) yields a reduction of 29.2 percentage points attributable to capacity-aware acceptance modeling. The acceptance-model contribution is roughly twenty times larger than the policy contribution, indicating that capacity-aware offer scoring is the dominant simulation-relevant lever in this setting.

The learned policy identifies risk-and-blood-type matching as the dominant compatibility axis.

Although the policy’s contribution to the discard rate is modest in absolute terms, the structure of the trained transformation matrix is independently informative about how the allocation problem decomposes within the medically compatible candidate pool. The trained policy departs substantially from its identity initialization (relative Frobenius norm $\|A - I\|_F / \|I\|_F = 0.92$), indicating that the policy gradient carried consistent information about which feature directions to emphasize even when the per-episode reward

signal was noisy. The reweighting is asymmetric and economically interpretable: the risk-axis feature (CPRA for patients, KDPI for donors) receives a column norm of 3.13, more than triple its initial weight, while the body-mass and creatinine dimension is downweighted to 0.82. ABO sub-categorization is also upweighted, with column norms of 2.69 for O, 2.44 for A, and 2.14 for age, but only 1.02 for the rare type B. The singular value decomposition of the trained matrix shows that all seven singular values exceed five percent of the largest, confirming that the policy uses the full dimensionality of the feature space rather than collapsing onto a low-rank subspace. The dominant singular direction loads jointly on risk and the A blood-type indicator, suggesting that the policy identified an interaction between organ risk and ABO sub-category as the most informative axis for differentiating among medically compatible candidates. These findings are substantive in their own right: they suggest that within the medically compatible pool, the binding constraint on which candidate to prioritize is not body-mass-based matching but risk-and-blood-type matching, and the asymmetric ABO weighting reflects the structural fact that finer matching has more scope to discriminate among many candidates in the common-blood-type categories than in the rare ones.

Capacity-acceptance coupling functions as designed. The offer-level Pearson correlation between predicted capacity probability and predicted acceptance probability in Arm 3 is -0.36 (Spearman -0.45 , $n = 4,152$), with mean predicted acceptance probability falling from 12.7% in the lowest-capacity-pressure quartile to 4.7% in the highest. This coupling is essentially identical in the RL-active and RL-inactive subsets, confirming that it is a property of the acceptance model itself rather than an artifact of the policy.

The capacity-aware model neutralizes capacity pressure as a driver of system-level discard. At the state level, the most rural states with substantial offer volume show simulated discard rates clustered around 4%, while urban states with high offer volume show rates exceeding 20% in several cases. The Pearson correlation between state rurality and

simulated discard rate strengthens from -0.12 in the full sample to -0.43 when restricted to states with at least 10 offers. The mechanism, established at the offer level above, is that the two-tower model assigns lower acceptance probabilities to offers at high-capacity-pressure centers, allowing the simulation to redirect organs toward lower-pressure centers downstream. The capacity-pressure correlation with state rurality, by contrast, is fragile under sample-size restrictions and should not be interpreted as evidence of a robust state-level rurality-capacity relationship.

6.2 Limitations

The findings reported above differ in their robustness, and the conclusion is best read with that hierarchy in mind. The strongest results in this paper come from the estimated structural models. The volume regression is identified on 18,732 center-month observations spanning eight years and produces large, well-determined coefficients on the surgeon-supply interaction and the metropolitan indicator. The acceptance GLMM, the donor-level discard model, and the capacity model are each estimated on samples of thousands of offers or hundreds of center-months, achieve held-out AUCs in the 0.82 to 0.86 range, and produce coefficients with clean economic interpretations. The three-way comparison among combination specifications is also well-powered, with AUC differences that are too large to plausibly reflect sampling variation, and the resulting interpretation about complementarity between match quality and surgical capacity rests directly on the structure of those estimates. These are the empirical findings that should carry the most weight in any reading of the paper.

The findings drawn from the simulation and from the reinforcement learning policy are more provisional. Two simulation results survive close scrutiny: the 29.2-percentage-point gap between Arm 2 and Arm 3 is large enough that even with the simulation’s simplifications it supports the qualitative claim that capacity-aware modeling is the dominant lever, and the offer-level Pearson correlation of -0.36 between predicted capacity probability and predicted acceptance probability is estimated on 4,152 offers and is therefore well-determined within

the simulation run. Other simulation outputs—the absolute discard rates produced by each arm, the policy contribution of -1.3 percentage points, the state-level rurality correlations—are based on a single calendar month of organ arrivals, a single random seed, and a stylized environment that abstracts from HLA matching, geographic prioritization, heterogeneous CIT accumulation, and other features of real allocation. These results should be read as directional evidence about how the components of the system interact rather than as estimates of magnitudes that would survive transport to a richer environment.

Several specific limitations qualify the simulation results in particular. First, the simulation is run on a single calendar month of organ arrivals (December 2022, 456 donor kidneys), which constrains both statistical power and external validity. Seasonal effects associated with late-year staffing constraints are visible in the real-world benchmark discard rate of 30.78%, somewhat above the long-run national average. Second, the simulation environment uses a permissive eligible candidate pool defined by blood-type compatibility within the active waitlist, abstracting away from the HLA matching, geographic proximity, and time-based prioritization rules of the actual allocation algorithm. The simulation also uses a deterministic CIT accumulation model that does not reflect the heterogeneous logistical delays driving a substantial share of real-world discards. Together these simplifications imply that simulated discard rates should not be interpreted as predictions for real-world deployment; the simulation is informative through between-arm comparisons rather than absolute outputs. Third, the metropolitan classification provided by SRTR contains only three non-metropolitan centers in the entire U.S. transplant system, severely limiting the statistical power of the center-level urban-rural analysis. Fourth, the calibration adjustment applied to the GLMM at simulation time corrects for the calibration-in-the-large component of the conditional-versus-marginal prediction problem documented by Pavlou et al. (2015) and Zeger et al. (1988), but does not implement the full slope correction; this is acceptable here because the simulation depends primarily on relative ranking of candidates, but a more complete correction would be required for applications relying on the absolute level of

predicted probabilities.

6.3 Future Work

Reinforcement learning for difficult-to-allocate organs. The learned policy in this report activates on 22.6% of December 2022 organs and produces a 1.3-percentage-point reduction in simulated discard rate when paired with capacity-aware acceptance modeling. This is a promising but modest effect, and the policy itself is parsimonious—a 7×7 transformation matrix over a small set of standardized features, trained with REINFORCE under a fixed offer-cap-based termination criterion. The combination of a meaningful learned structure (Section 5.10) and a small aggregate effect on discards indicates that the policy framework deserves substantially more attention than this report could give it. The most pressing avenue is reward function design. The current reward is binary on episode-level acceptance, which produces noisy gradient signal and incentivizes only finding *any* accepting candidate within the offer cap rather than finding the best one. Richer reward formulations would weight outcomes by life-years from transplant against the option value of waiting, by cumulative cold ischemia time accrued during the offer chain, by post-transplant graft survival, or by some combination of these—each of which is theoretically motivated and would change the policy’s incentives in economically meaningful ways. A dedicated paper centered specifically on the RL component would have room to explore these reward formulations systematically, evaluate richer policy architectures (recurrent or attention-based networks that condition on the full offer history rather than a fixed feature summary), implement value-function-based methods that estimate the option value of additional offers given the current chain, and conduct robustness checks against different gating thresholds, training windows, and termination criteria. The high-discard-risk subset of organs is the natural target for such a paper, since the marginal value of improved reranking is largest where the existing match-run order produces long failed offer chains that consume cold ischemia time without yielding acceptance; the present results establish the basic feasibility of the approach and

document the structural patterns the policy learns, but the design space of the policy itself is largely unexplored within this report.

Investigating the metropolitan-rural throughput gap beyond surgeon count. The volume regression identifies a +2.023 transplant-per-month metropolitan advantage that persists after conditioning on surgeon supply, offer volume, mean organ quality, and cold ischemia time. This unexplained gap is substantial—roughly one-third of the sample-mean monthly throughput—and suggests structural factors operating beyond the surgical workforce channel that the present analysis cannot identify. Promising avenues include (i) explicit modeling of referral networks and the pipeline through which patients arrive at transplant centers, drawing on the “selective referral” framework of Sonnenberg et al. (2019); (ii) finer-grained measurement of clinical infrastructure, including subspecialty consult availability, multidisciplinary care team composition, and access to specialized perioperative resources; (iii) examination of the interaction between center metropolitan status and patient socioeconomic characteristics, since rural patients face documented barriers to listing and post-transplant care (Axelrod et al. 2008) that may compound center-level differences; and (iv) longitudinal analysis tracking how the metropolitan advantage evolves as centers grow or contract their surgical staff, which would distinguish persistent infrastructure effects from transient workforce shocks. Identifying which mechanisms drive the unexplained metropolitan gap is necessary for any policy intervention that seeks to expand transplant capacity in non-metropolitan settings.

Multi-period simulation and external validity. Extending the simulation beyond a single calendar month would strengthen the external validity of the between-arm comparisons. A multi-month or multi-year evaluation window would average over seasonal effects, expose policies to the full range of donor and waitlist conditions, and permit estimation of standard errors on simulation outcomes through reseeding rather than asymptotic approximations. A natural extension would also incorporate the survival outcome model from

Agarwal et al. (2025) alongside the acceptance and discard models used here, allowing policy evaluation against life-years-from-transplant rather than discard rates alone.

6.4 Closing Remarks

The strongest empirical contributions of this paper are the estimated structural models. The volume regression establishes that surgeon supply imposes a binding capacity constraint on transplant throughput at the center-month level and identifies a substantial metropolitan-rural gap that operates beyond the surgeon-supply channel. The acceptance, discard, and capacity models produce well-calibrated, interpretable predictions on held-out data and support a coherent decomposition of offer acceptance into match-quality and capacity-driven channels. The three-way comparison among combination specifications, in turn, suggests that match quality and surgical capacity behave as complementary inputs into the acceptance decision rather than as substitutes that trade off at a constant rate. These structural findings rest on samples of thousands of offers or tens of thousands of center-months and are the most secure foundation on which subsequent work can build.

The simulation and reinforcement learning components extend these structural results into a counterfactual analysis of allocation outcomes, but their conclusions are more provisional. Within the simulated environment, capacity-aware acceptance modeling is the dominant lever and the learned policy produces a smaller but directionally favorable reduction in discards. The structure the policy learned is itself informative: risk-and-blood-type matching emerges as the dominant compatibility axis within the medically compatible pool, and body-mass-based matching is downweighted. These findings document the basic feasibility of the approach and provide a structural reading of the candidate pool, but they do not pin down magnitudes that would survive transport to a richer simulation environment with HLA matching, geographic prioritization, and multiple calendar months of organ arrivals.

The most pressing extensions follow from this hierarchy. A dedicated treatment of the reinforcement-learning policy framework—particularly reward function design and richer pol-

icy architectures—would convert the directional evidence here into more decisive results, and investigation of the metropolitan-rural gap that the volume regression identifies would explain a structural pattern that the present analysis can document but not decompose.

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